

RL과 ICL로 수능 수학 정복하기

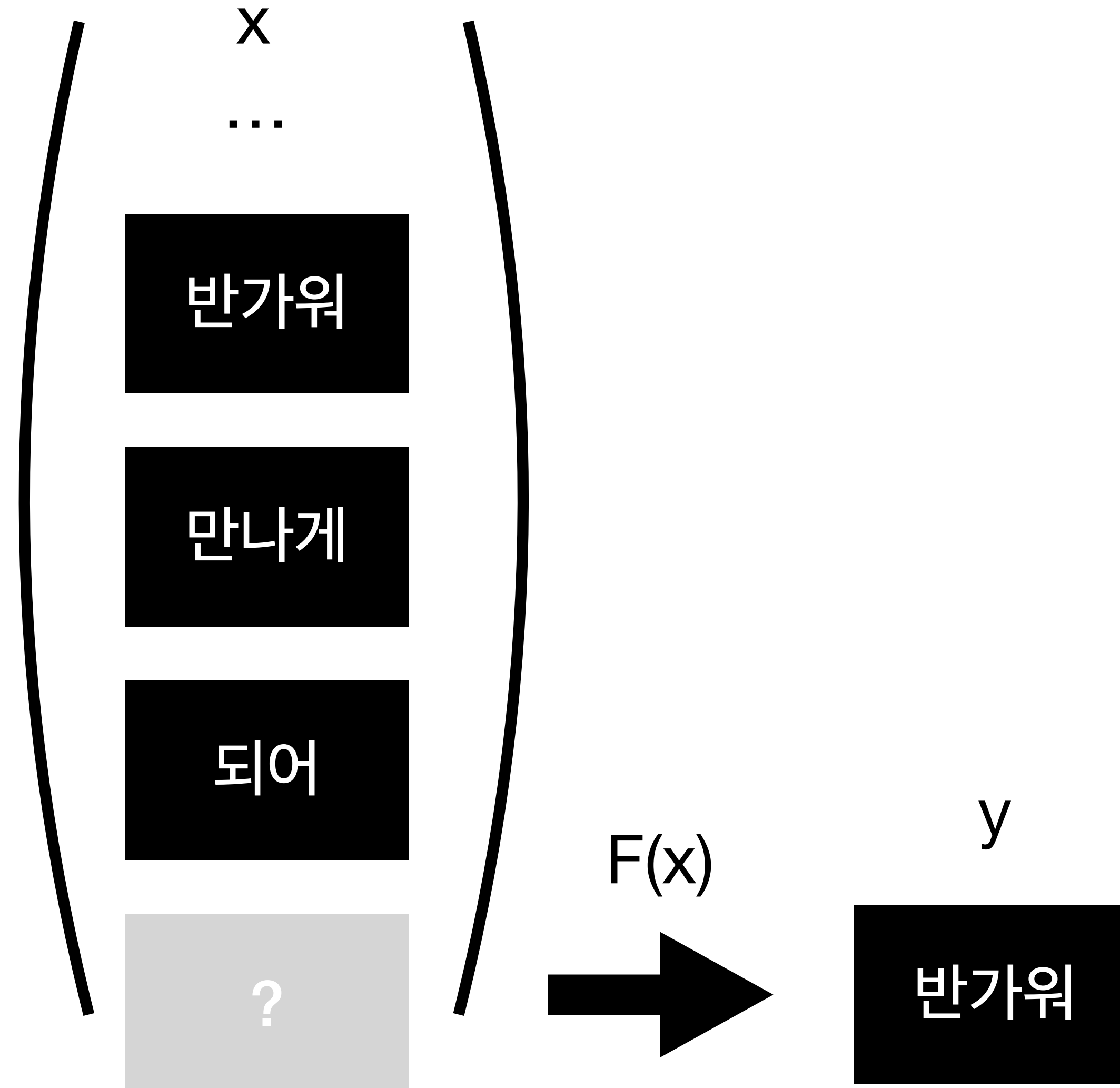
- 정말 할 수 있을까요? -

2장 : Transformer Anatomy

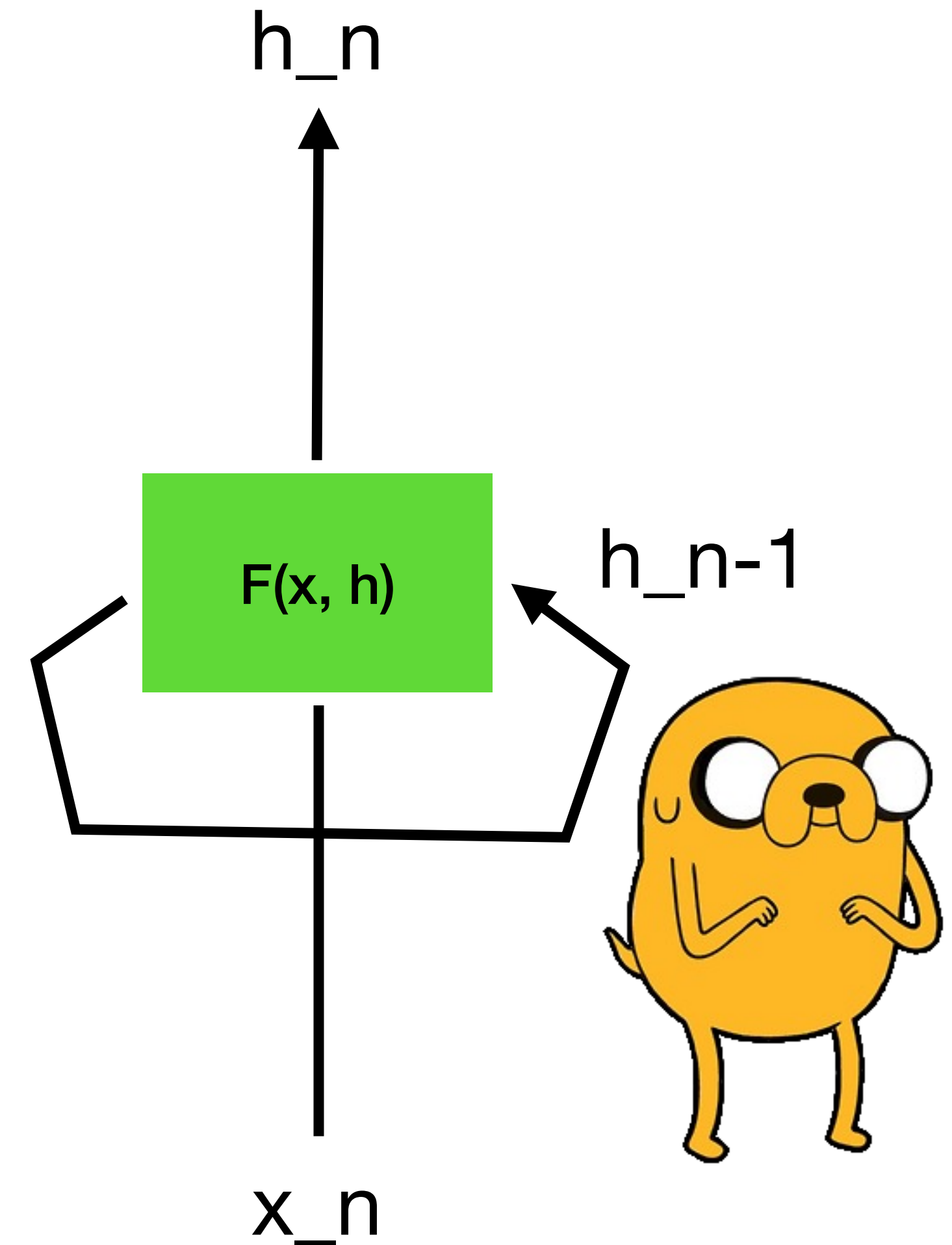
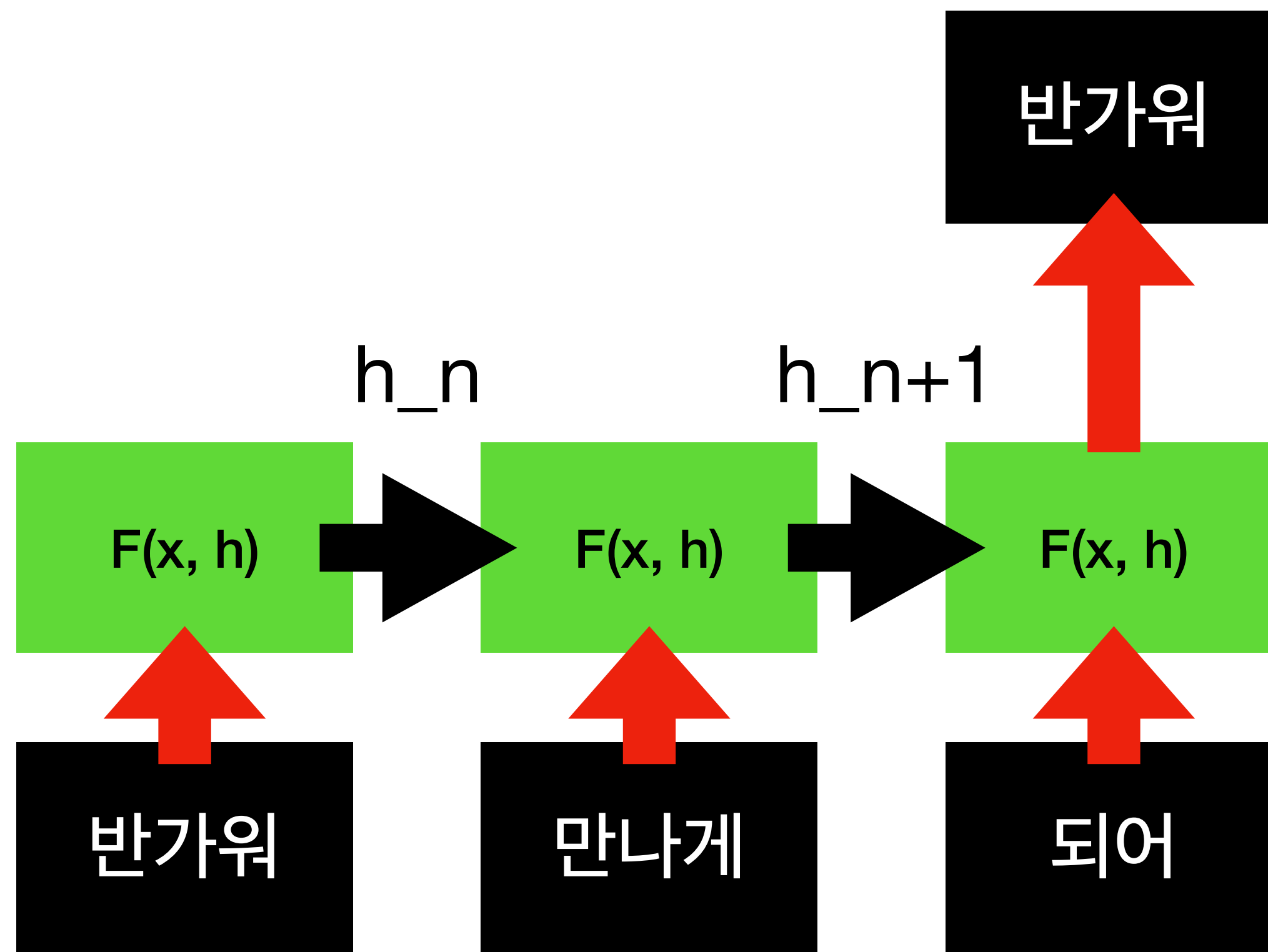
Todos...

- 트랜스포머 해부하기(...)
- 과제 안내하기(fun!)

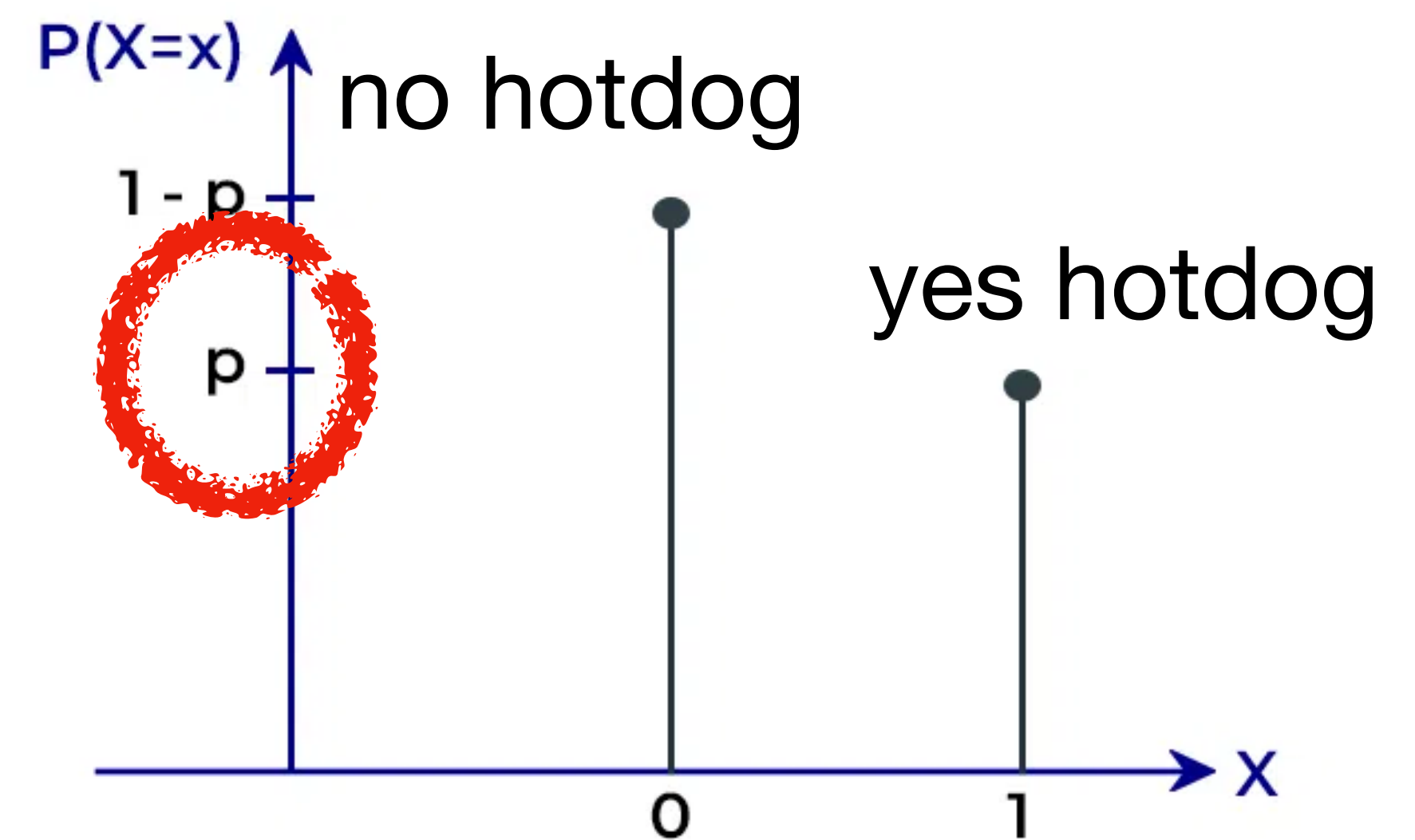
find $F(x)$?



find $F(x)$? - rnn



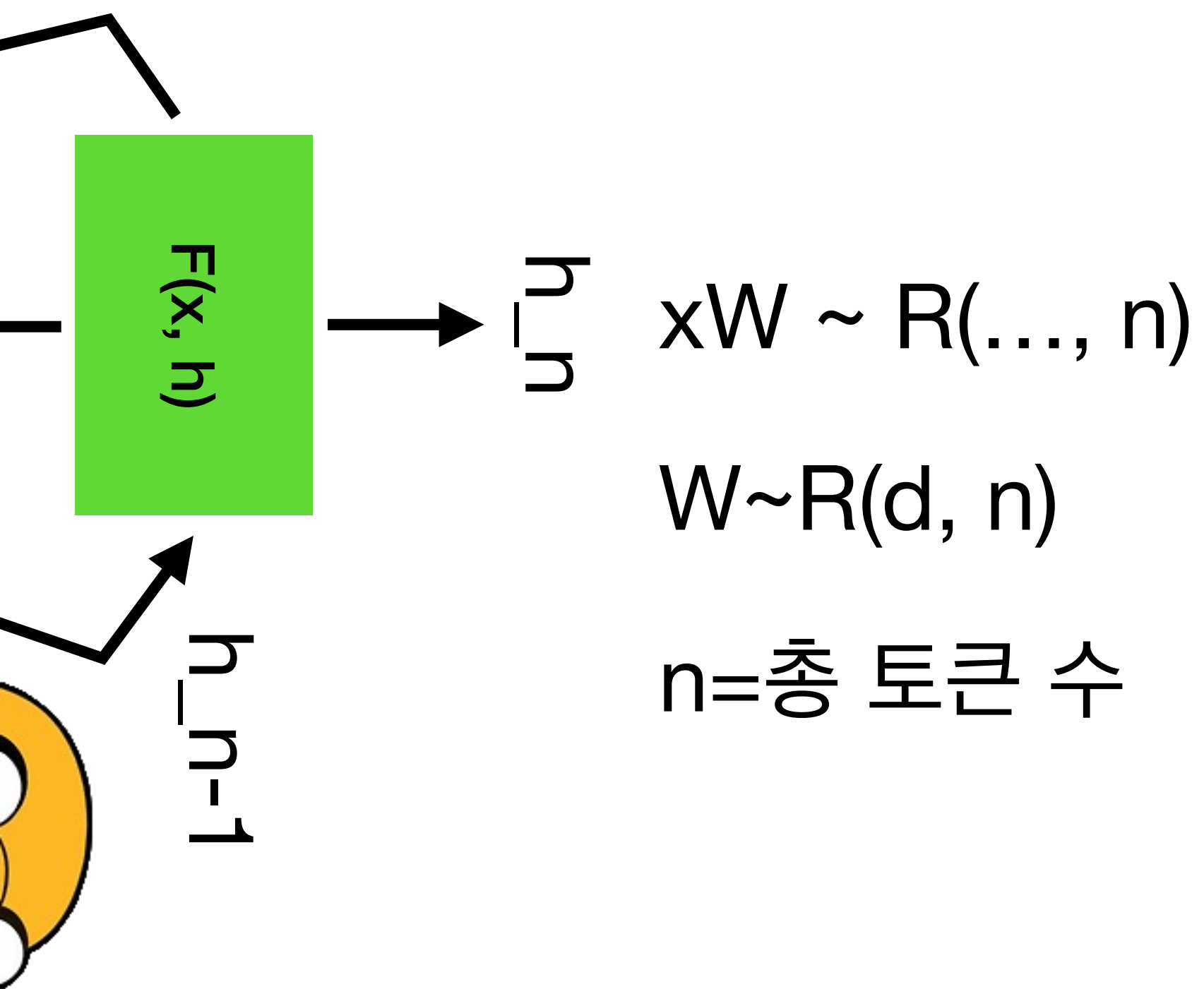
re: 인공지능은 뭘 배우냐?



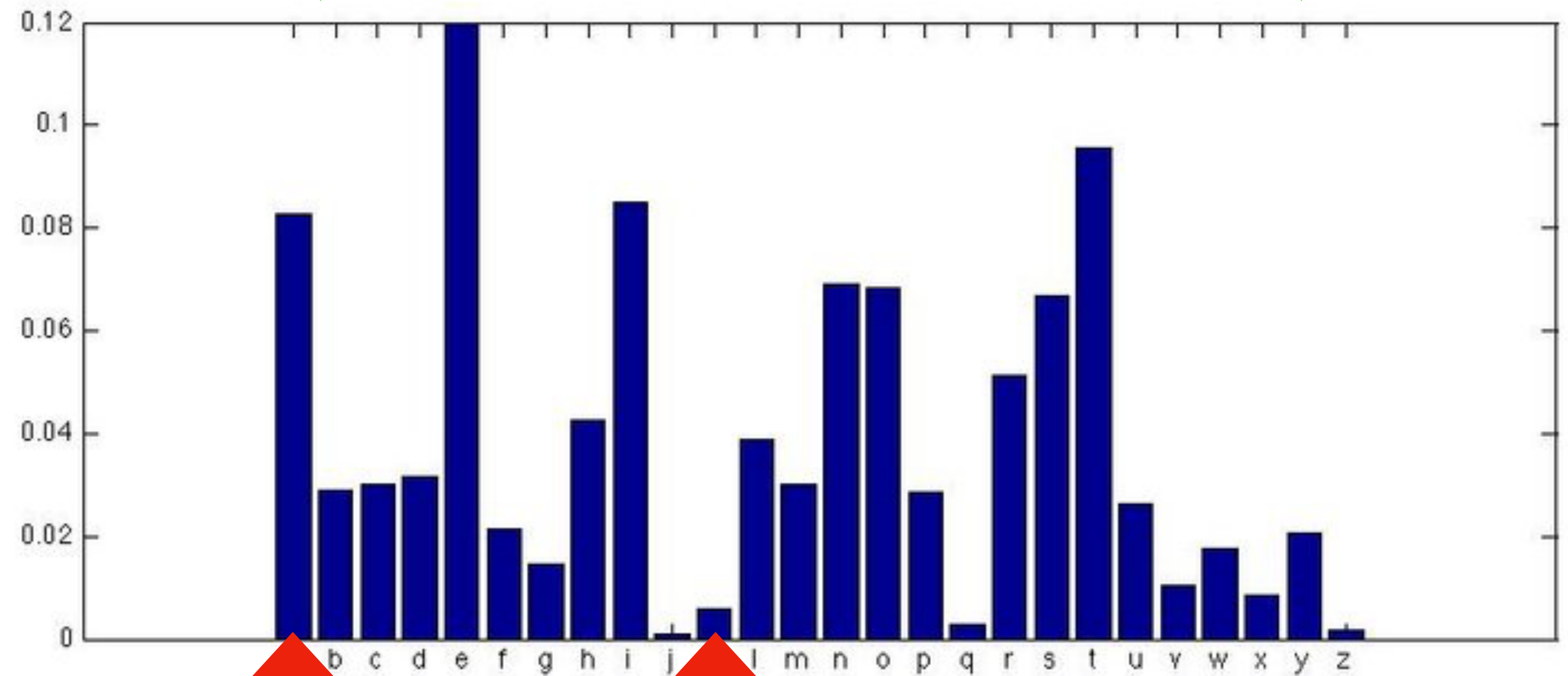
베르누이 분포

$$P(y|w) = p^y (1-p)^{1-y}, y \in 0, 1$$

find $F(x)$? - rnn



소프트맥스 후 n 개 토큰별 확률 분포



만나게

반가워

find F(x)? - rnn

$xW \sim R(\dots, n)$

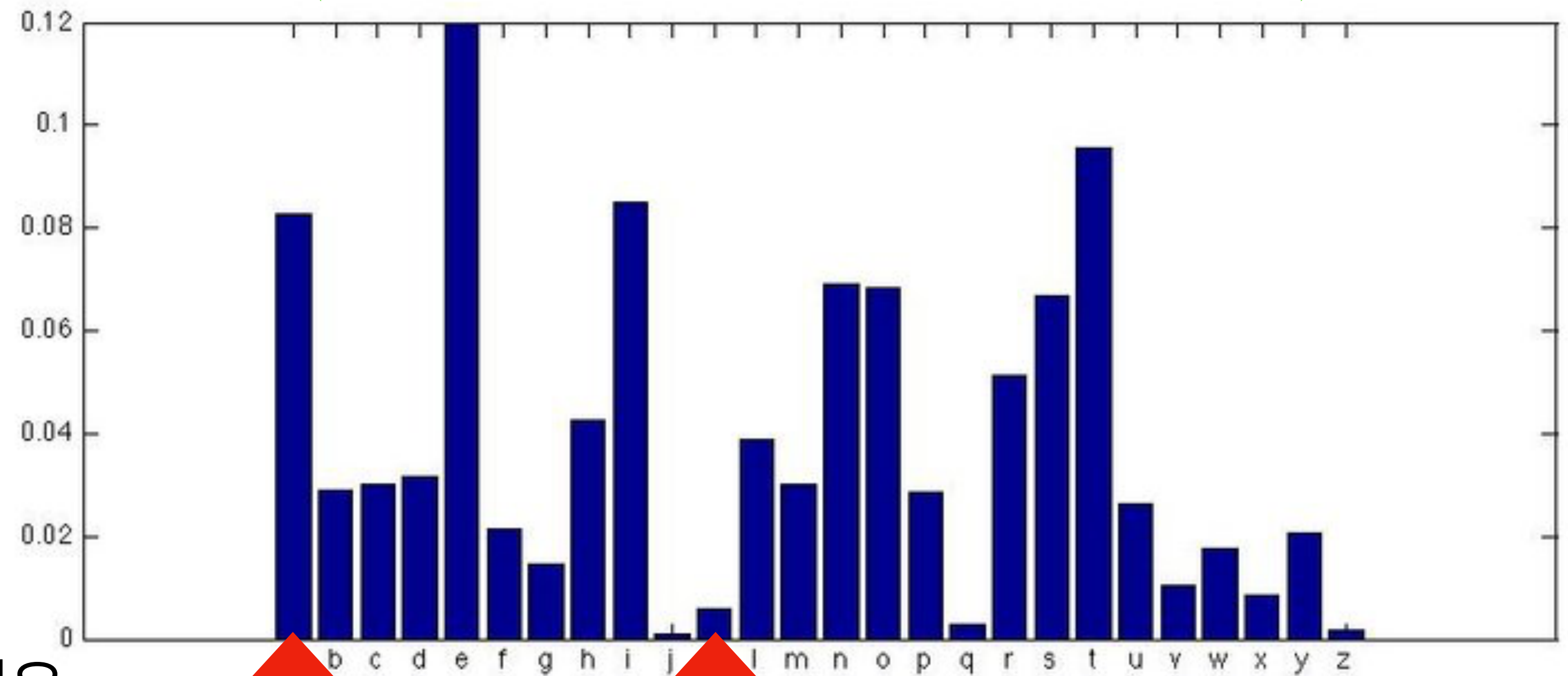
$$s(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

여기에 e가 등장하는 이유!

~~sigmoid가 activation이 되는 이유

~~boltzman distribution!

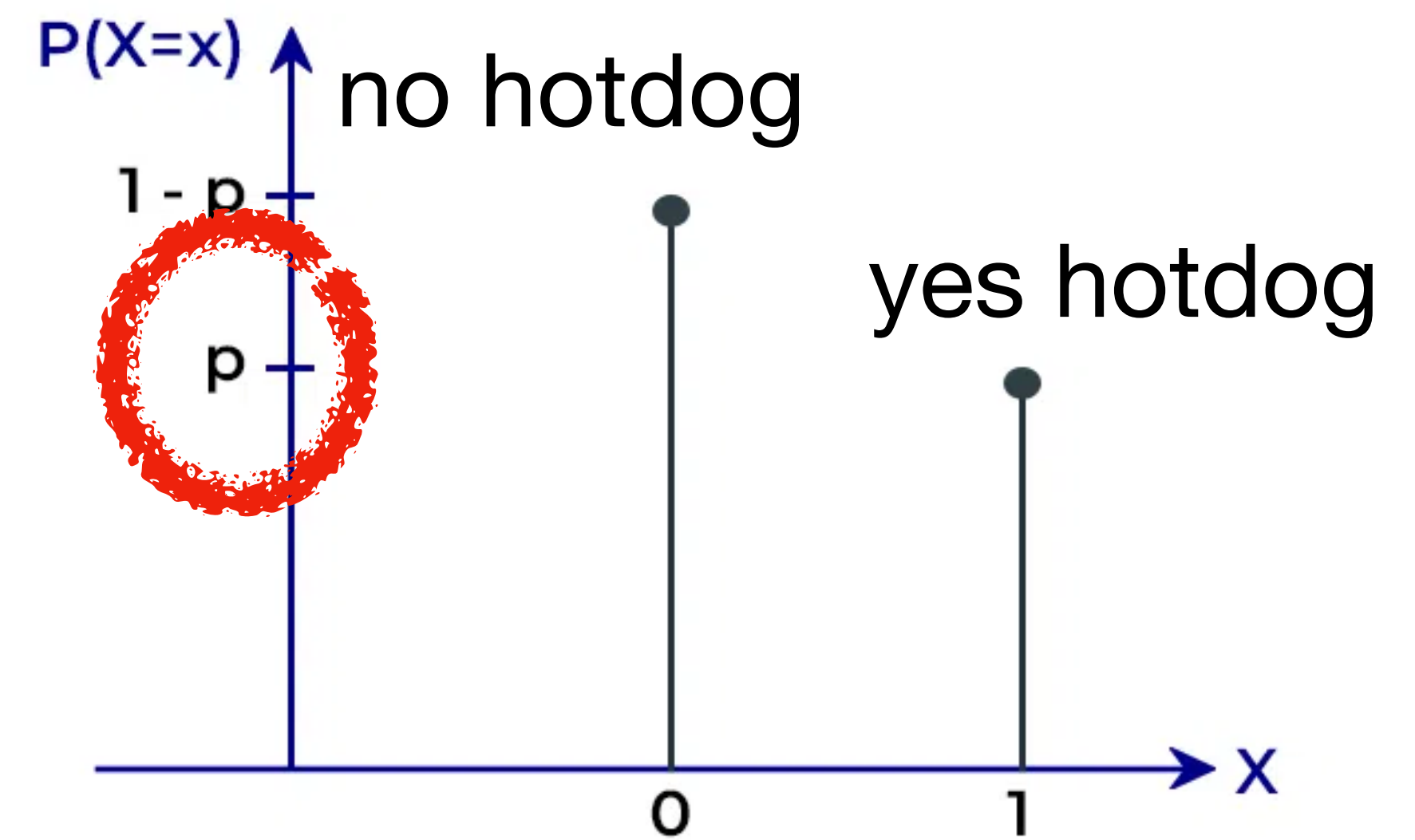
소프트맥스 후 n개 토큰별 확률 분포



만나게

반가워

re: 인공지능은 뭘 배우냐?

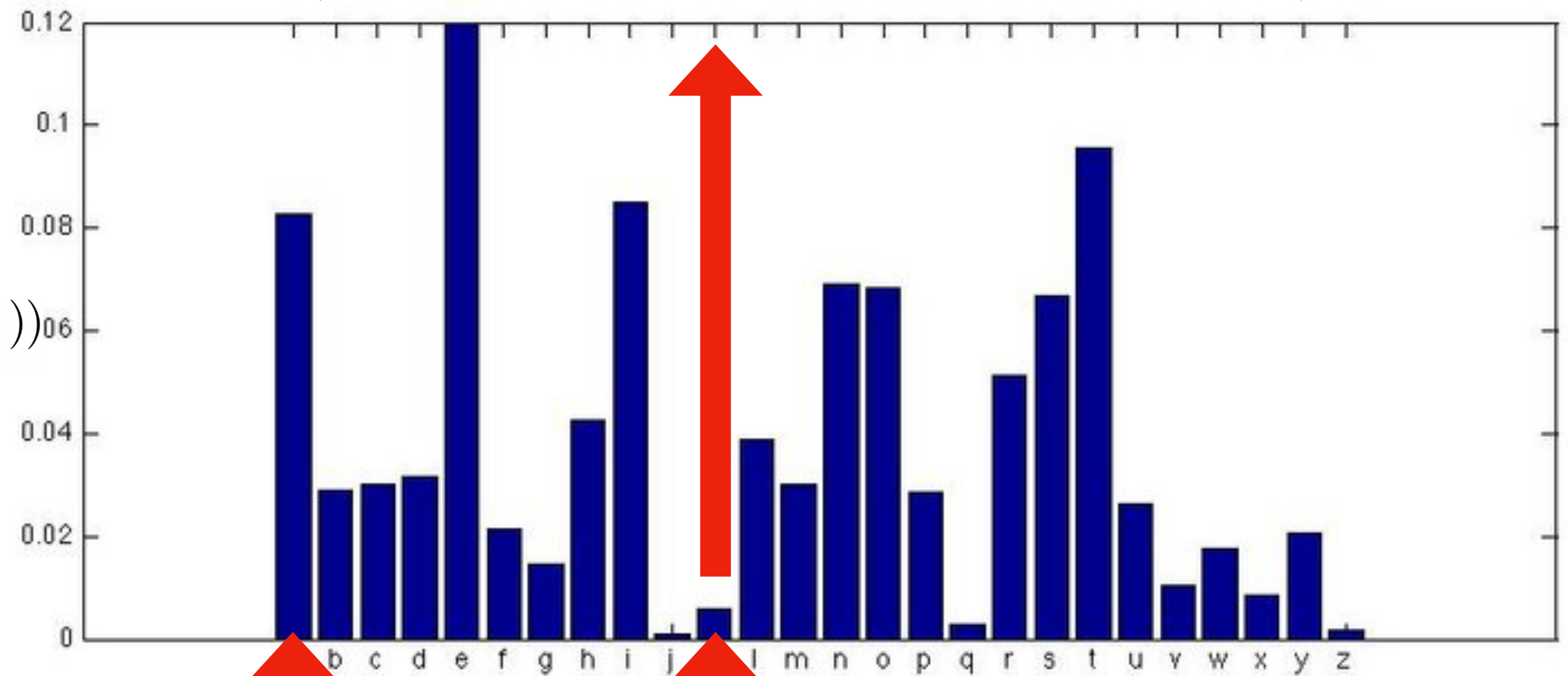


베르누이 분포

$$P(y|w) = p^y (1-p)^{1-y}, y \in 0, 1$$

find F(x)? - rnn

소프트맥스 후 n개 토큰별 확률 분포



만나게

반가워

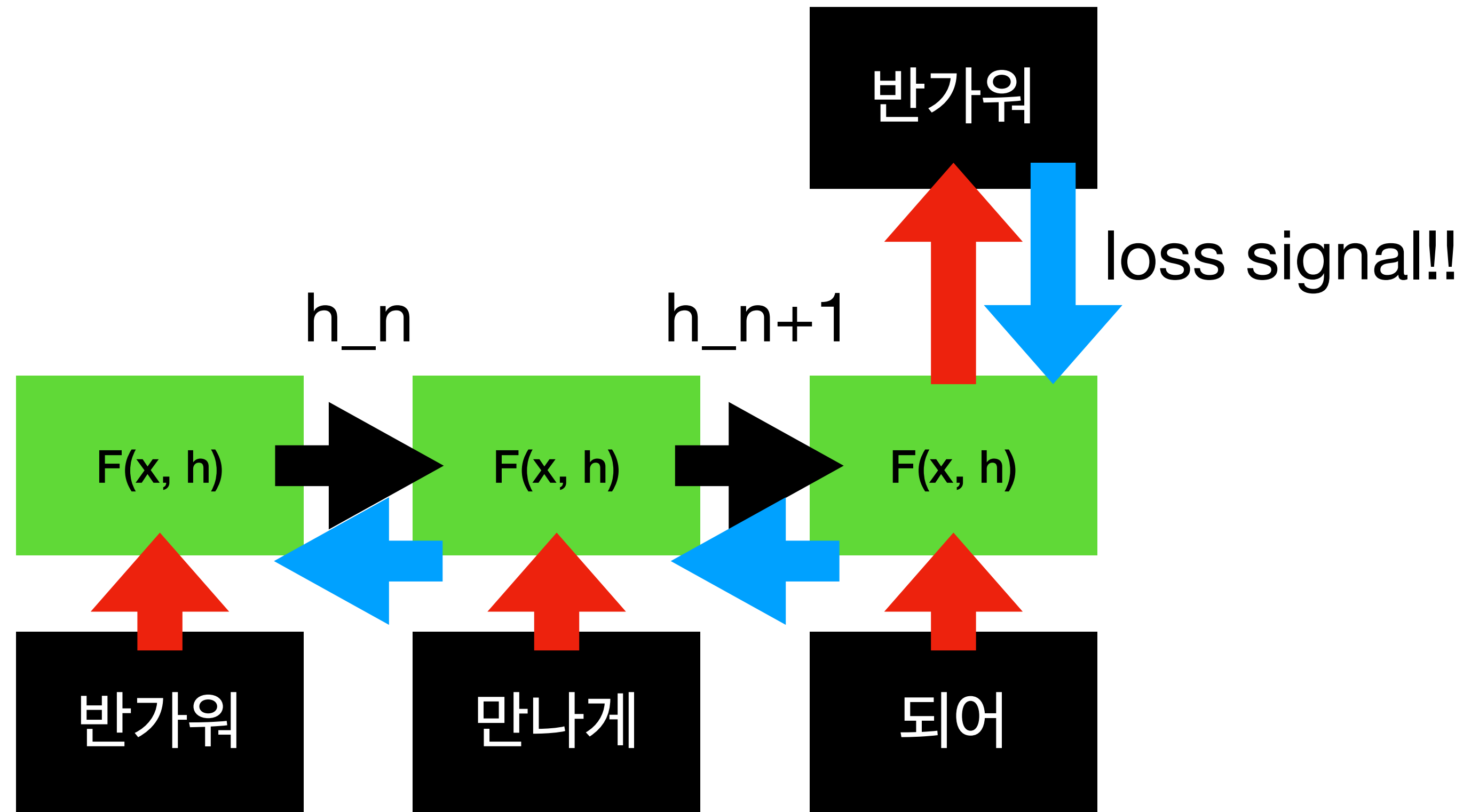
$$\operatorname{argmax}_{\theta}(\log(p_{\text{반가워}})) = \operatorname{argmin}_{\theta}(-\log(p_{\text{반가워}}))$$

loss signal!!

find F(x)? - rnn

$$\operatorname{argmax}_{\theta}(\log(p_{\text{반가워}})) = \operatorname{argmin}_{\theta}(-\log(p_{\text{반가워}}))$$

loss signal!!

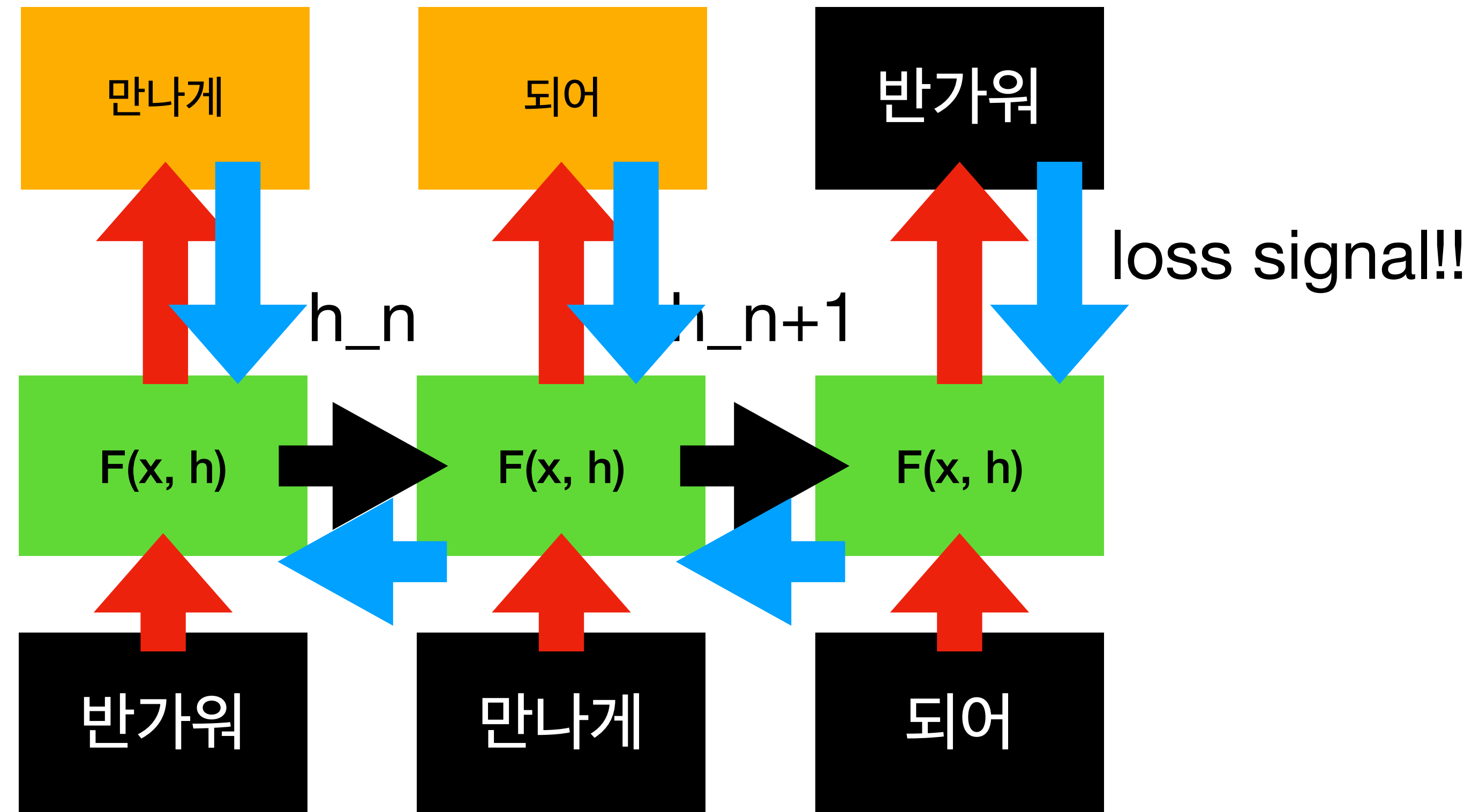


find F(x)? - rnn

$$\operatorname{argmax}_{\theta}(\log(p_{\text{반가워}})) = \operatorname{argmin}_{\theta}(-\log(p_{\text{반가워}}))$$

loss signal!!

teacher forcing method



find $F(x)$? - why rnn cool

- 같은 함수의 재사용(=infinity 를 length)
- hidden state를 통한 문맥의 반영

find $F(x)$? - why rnn sucks

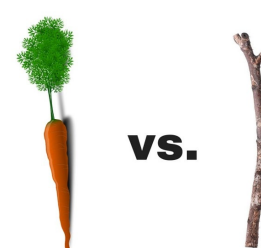
- hidden state를 통해 전달되는
gradient flow의 소실
- hidden state 특정 정보가 지금 필요 없다고 계속 필요 없지는 않다.

이것만으로는 부족하다..!

$$\frac{dLoss}{dx_1} = \frac{dLoss}{dx_{n-1}} \times \dots \times \frac{dx_2}{dx_1}$$



- 모두 1 초과이면 발산!
- 모두 1 미만이면 0

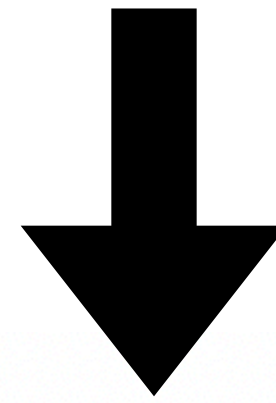


VS.



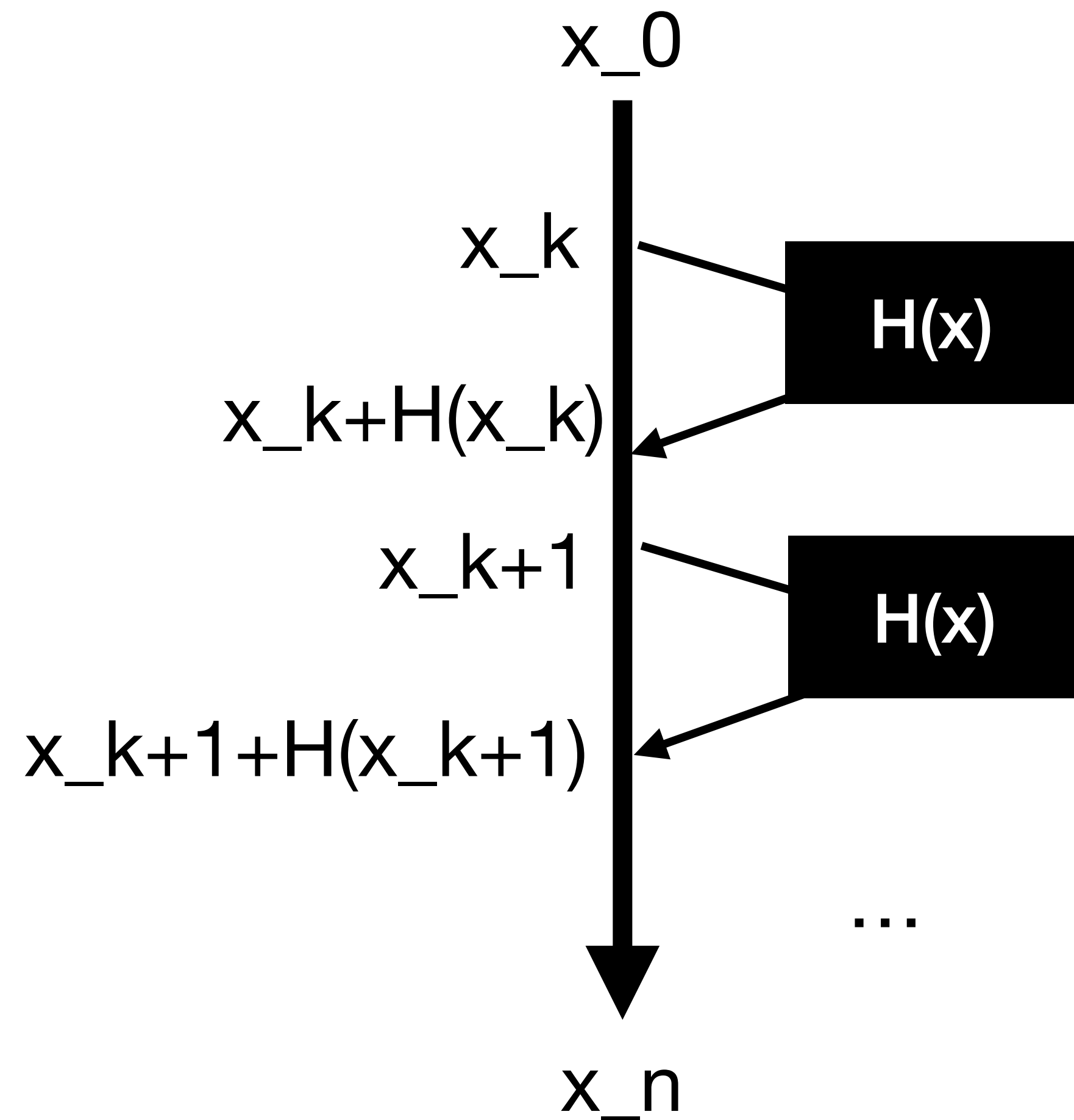
1번 문제에 대하여 - residual mapping

$$x_{n+1} = F(x_n)$$

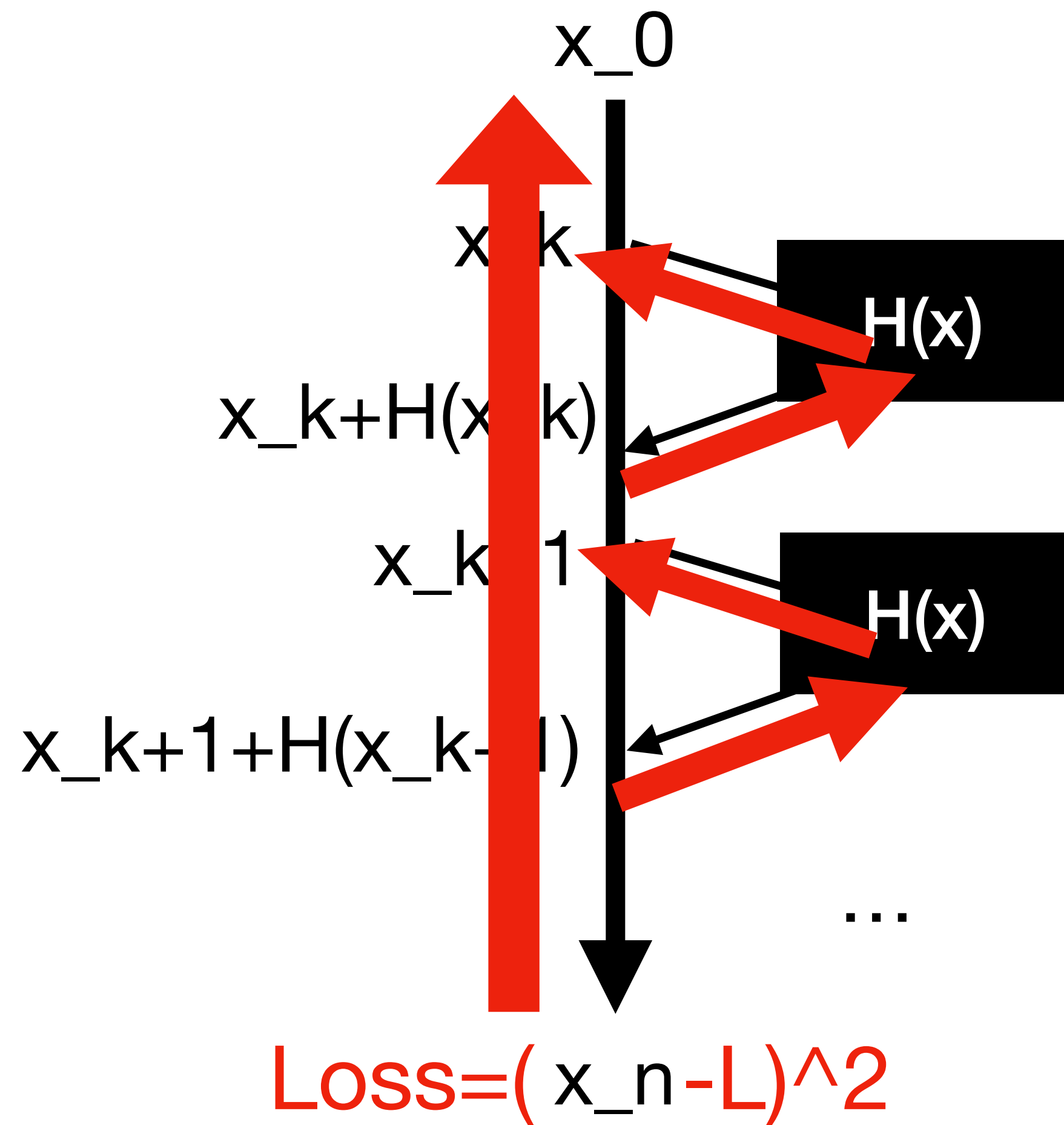


$$x_{n+1} = x_n + H(x_n),$$
$$H(x_n) = x_{n+1} - x_n$$

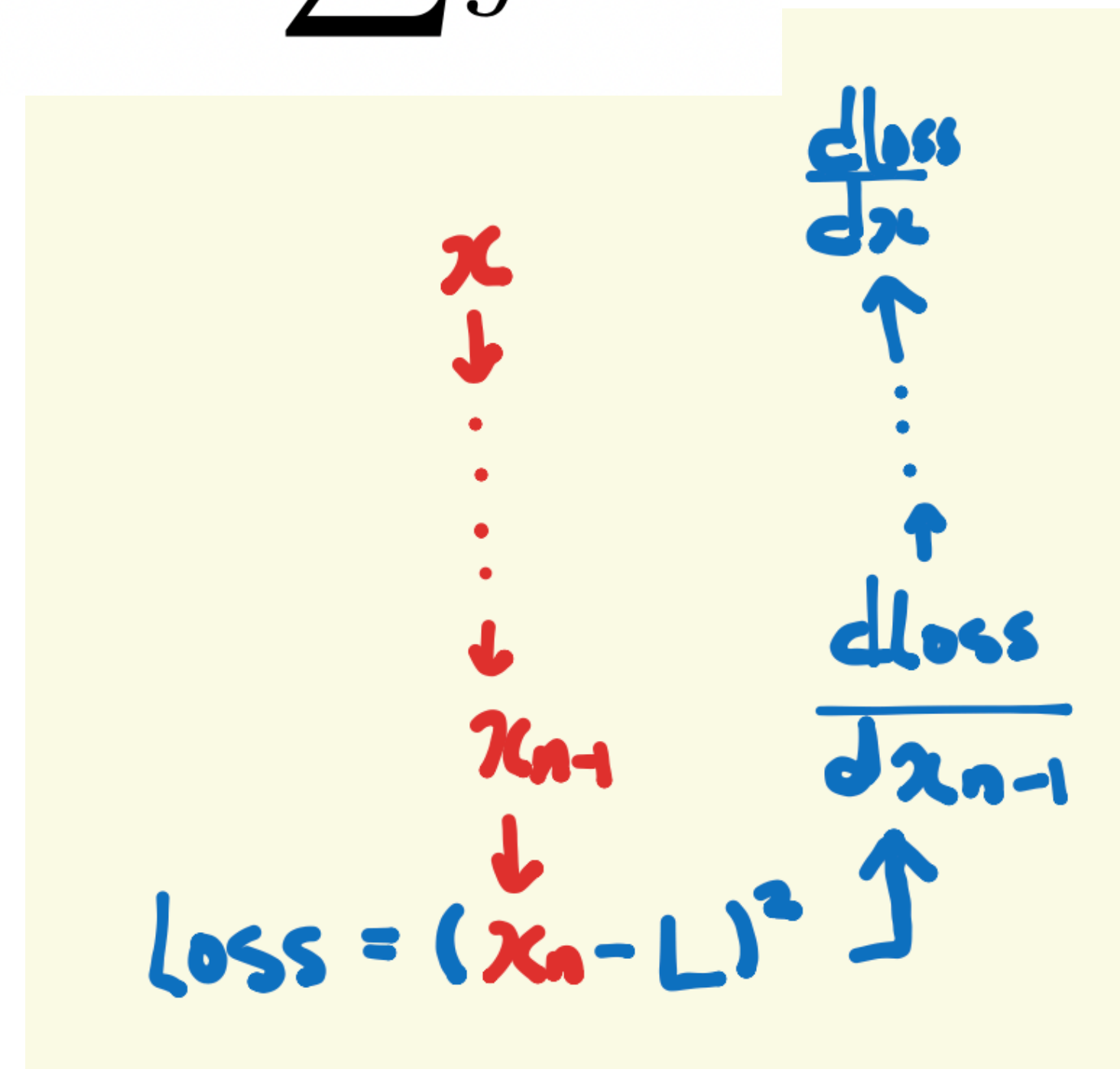
1번 문제에 대하여 - residual mapping



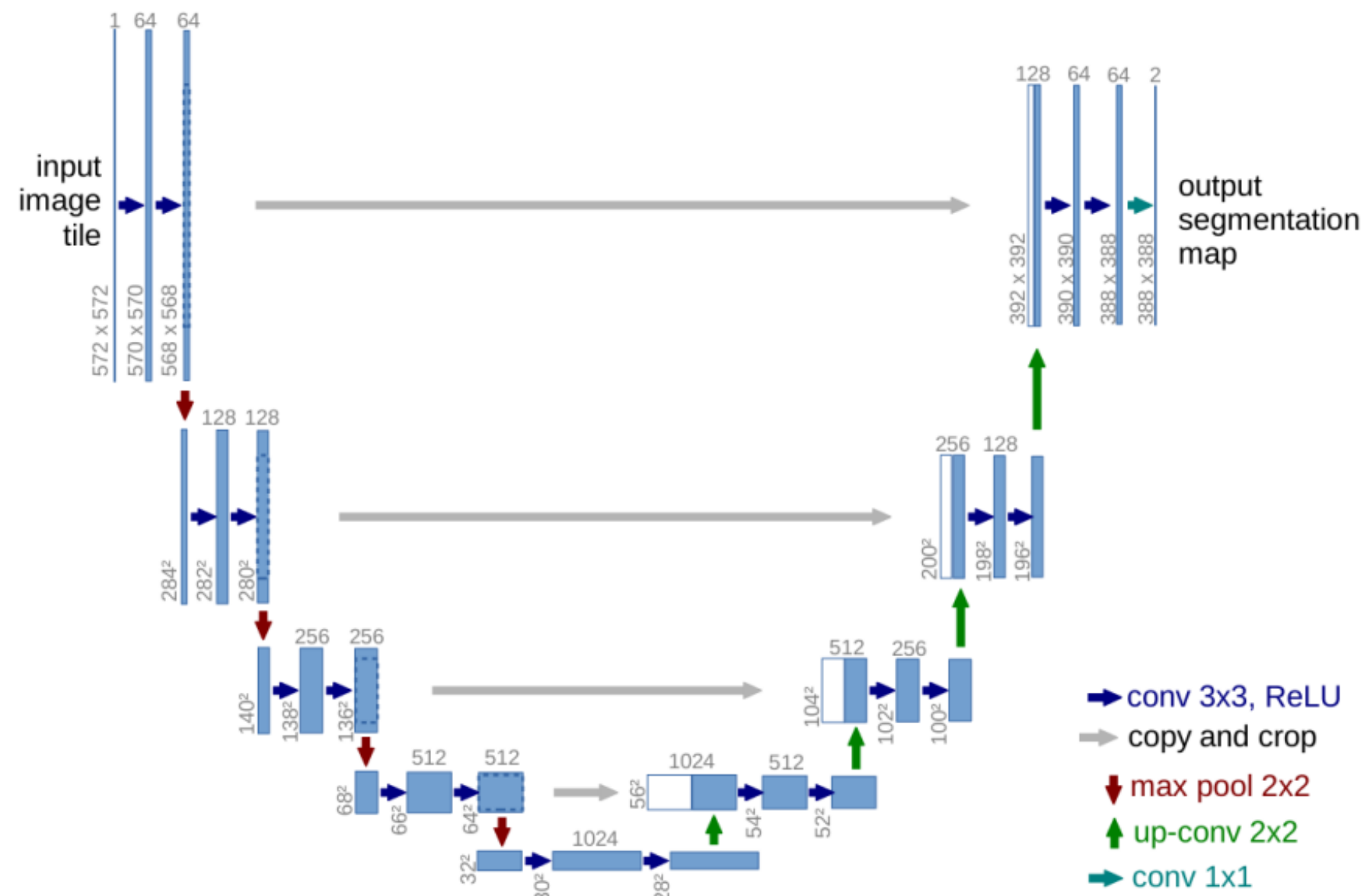
1번 문제에 대하여 - residual mapping



$$\prod grad \rightarrow \sum grad$$

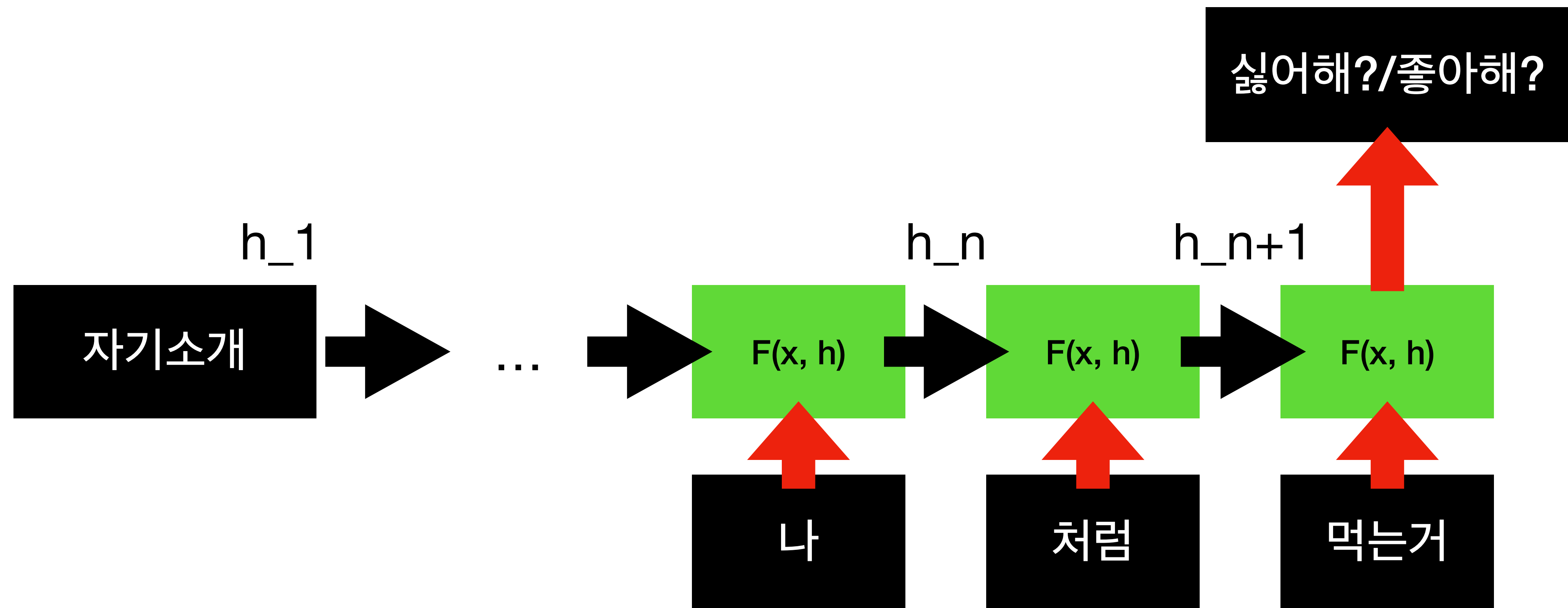


1번 문제에 대하여 - residual mapping

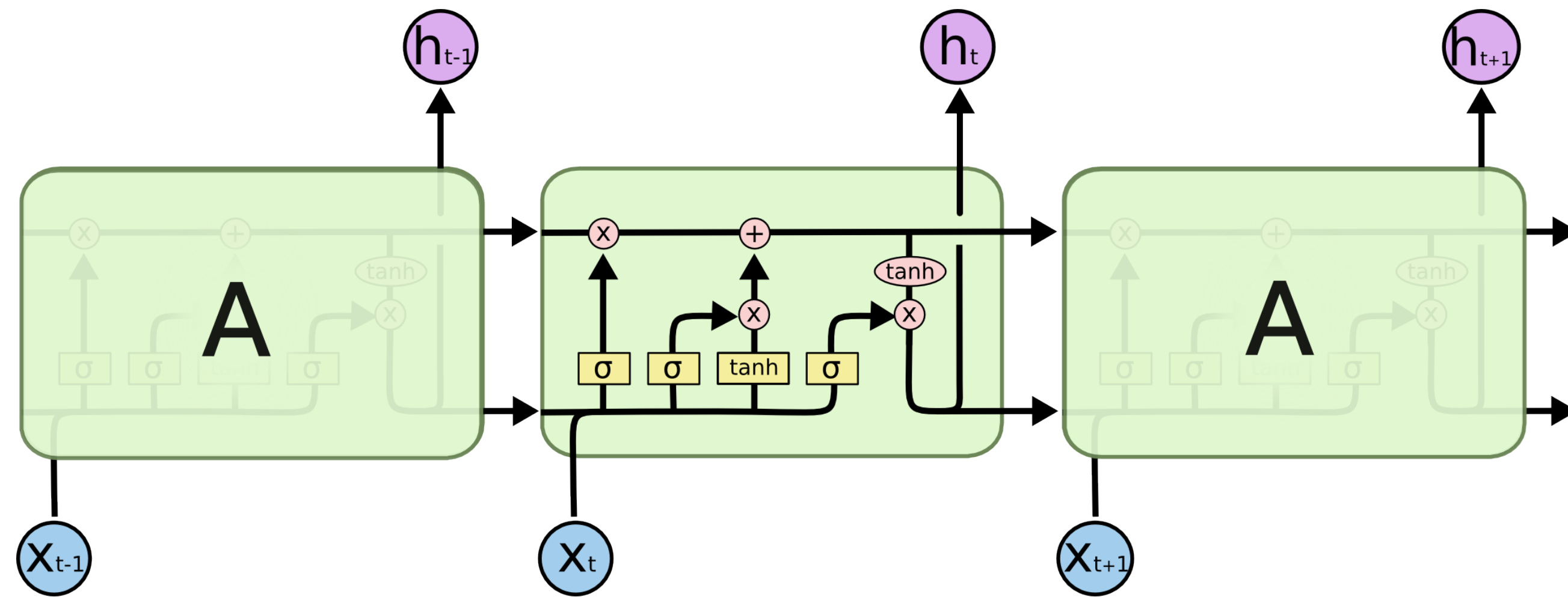


U-Net: Convolutional Networks for Biomedical Image Segmentation

2번 문제에 대하여



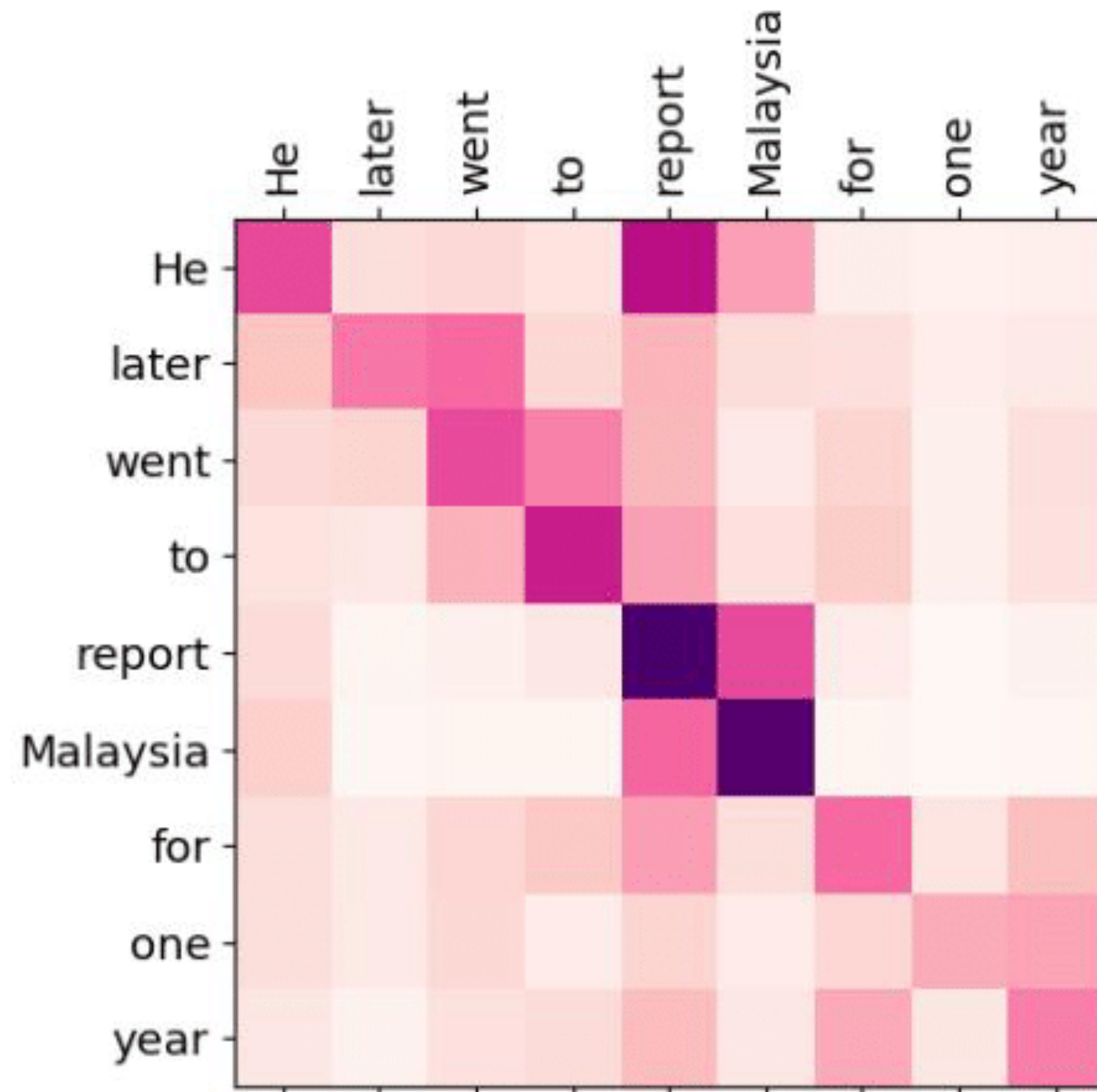
2번 문제에 대하여



LSTM, GRU-global information(무엇을 기억할지 무엇을 잊을지)
(<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>)

+인공지능이 어떻게 잘 배우게 함?

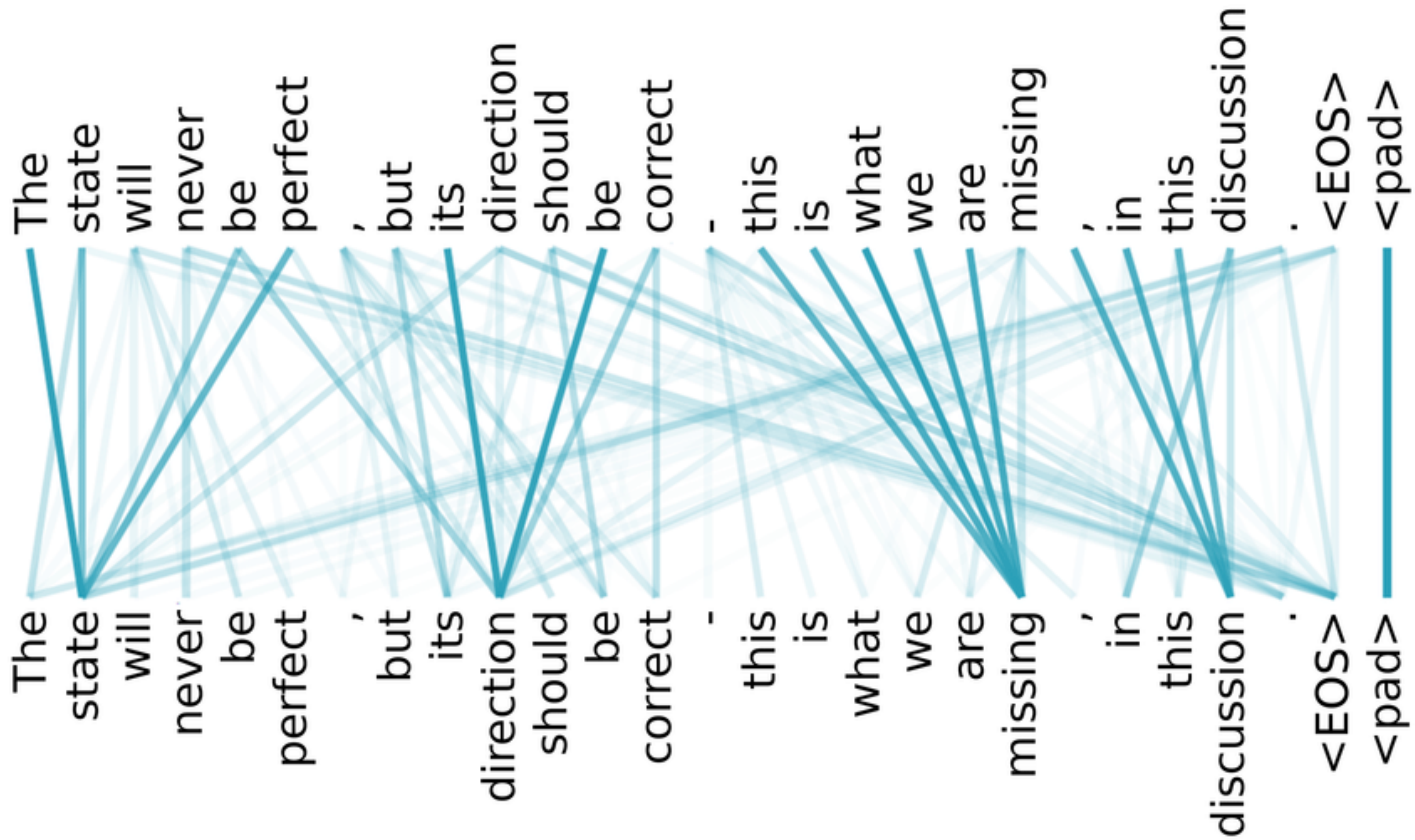
아키텍처의 끝판왕 = Transformer



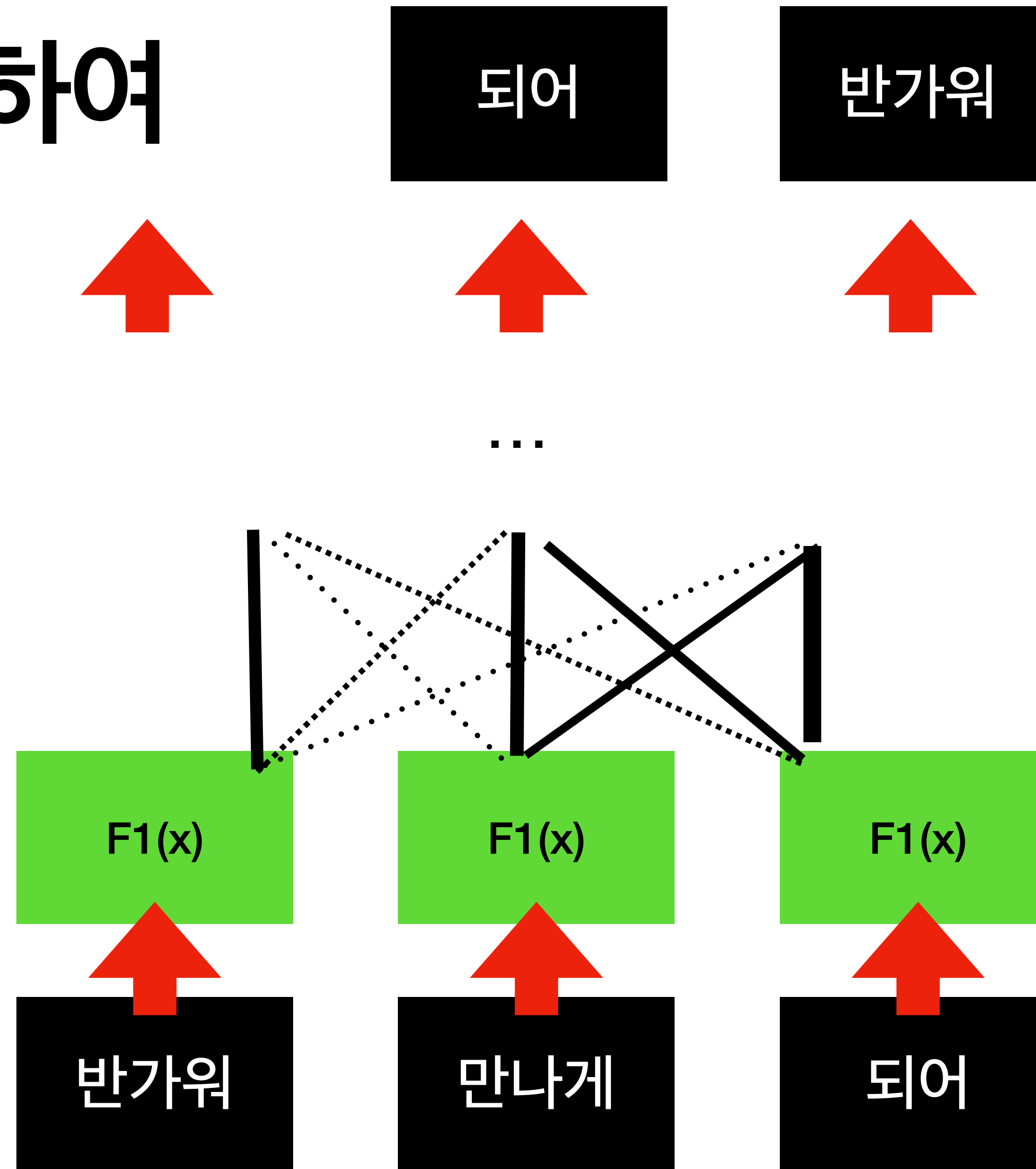
내적 = 두 친구 사이의 유사도



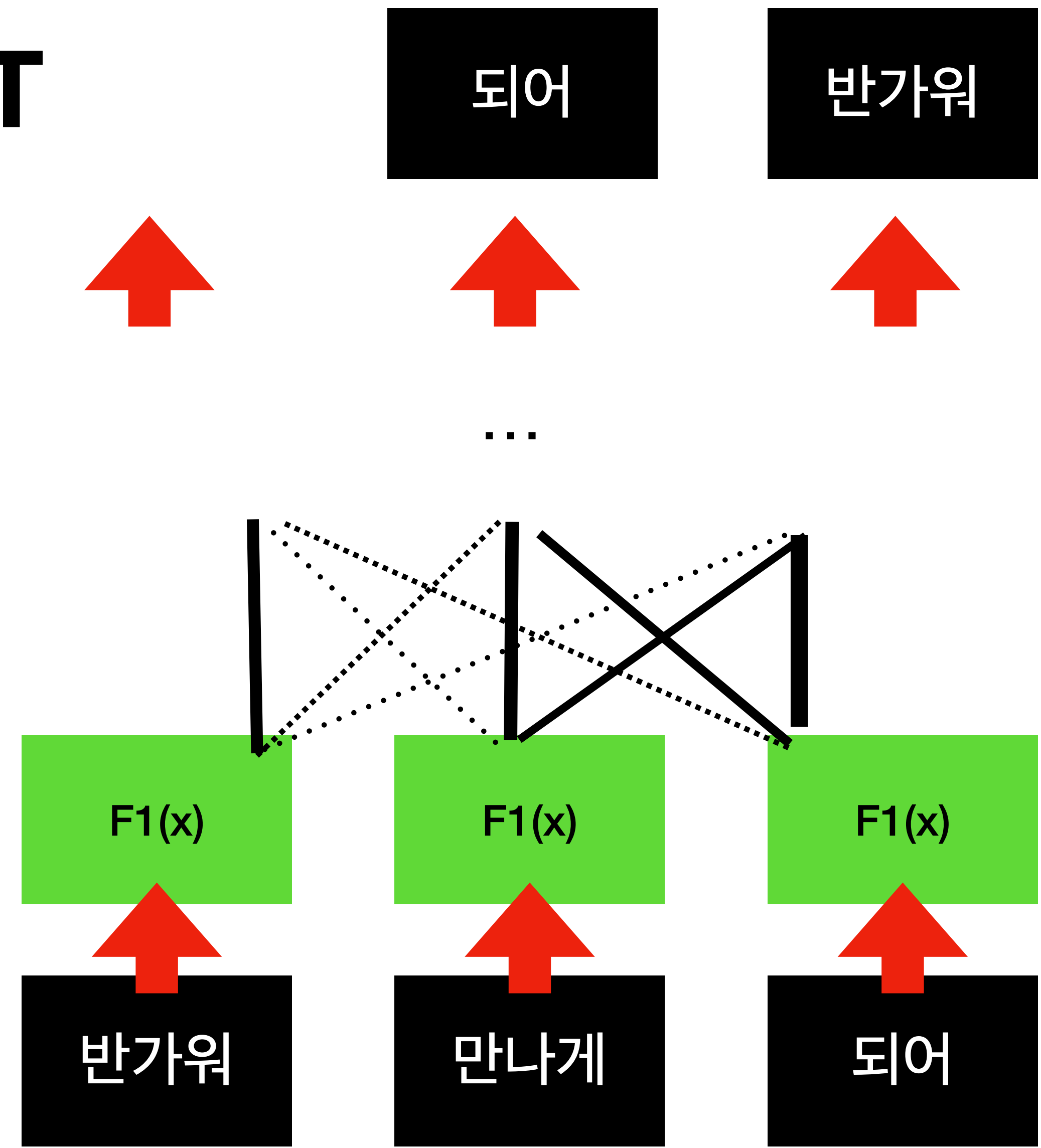
데이터를 처리할 때
어디에 집중해야 할지를 제시



2번 문제에 대하여 -GPT



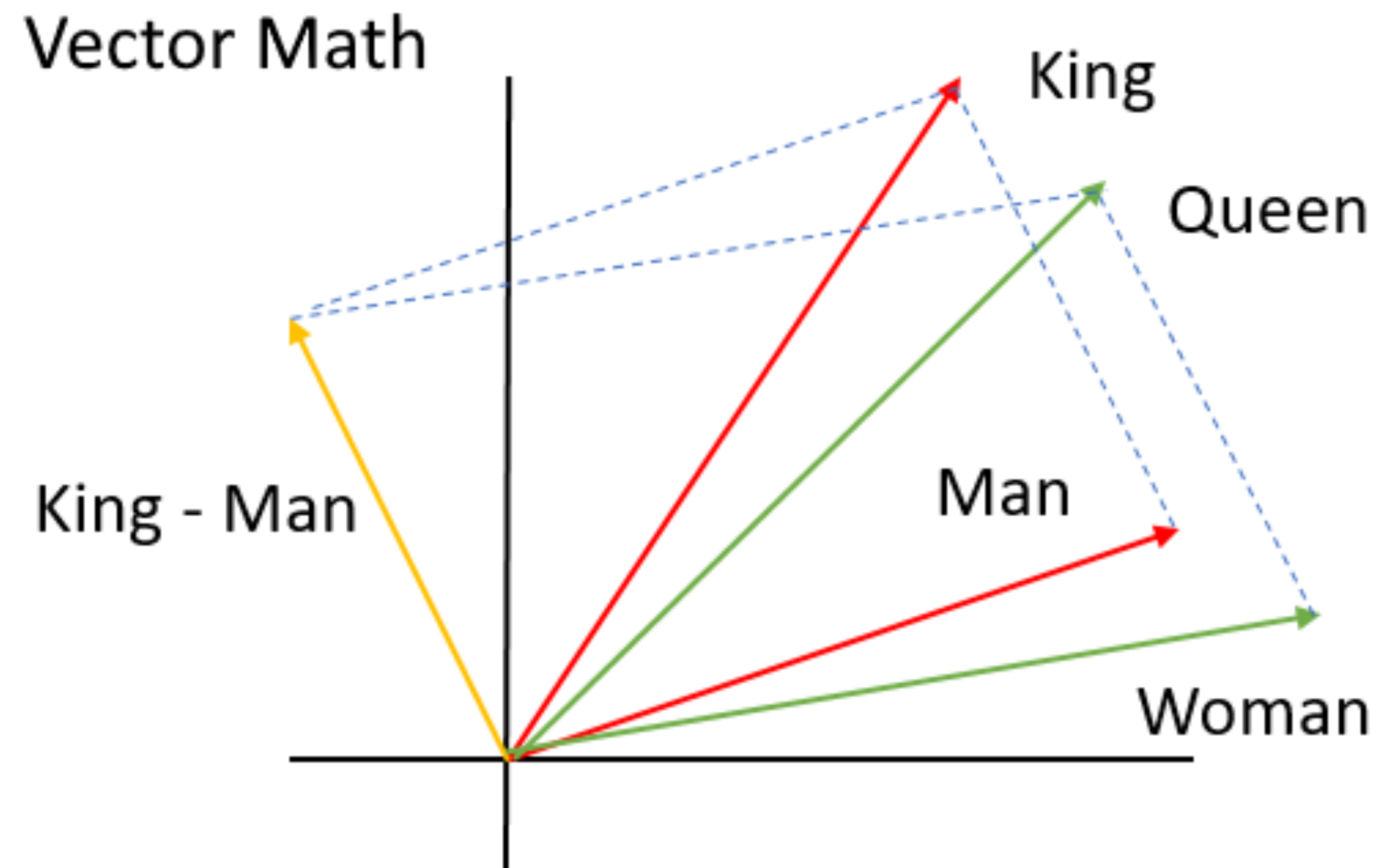
find $F(x)=GPT$



+인공지능이 어떻게 잘 배우게 함?

Embedding 임베딩

안녕 -> [0, 2, 6, 3, 1, 5]

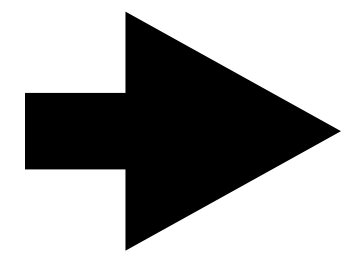


왕 - 남자 + 여자 => 여왕

문자란 무엇일까? for representing as vector

```
[1]: text = "안녕하세요"  
text.encode("utf-8")
```

```
[1]: b'\xec\x95\x88\xeb\x85\x95\xed\x95\x98\xec\x84\xb8\xec\x9a\x94'
```



```
[8]: int("0xec", 16)
```

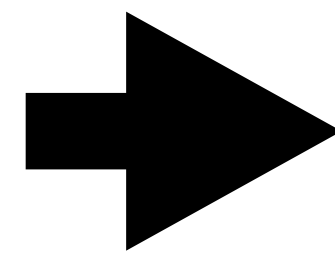
```
[8]: 236
```

```
[9]: int("0x95", 16)
```

```
[9]: 149
```

```
[10]: int("0x88", 16)
```

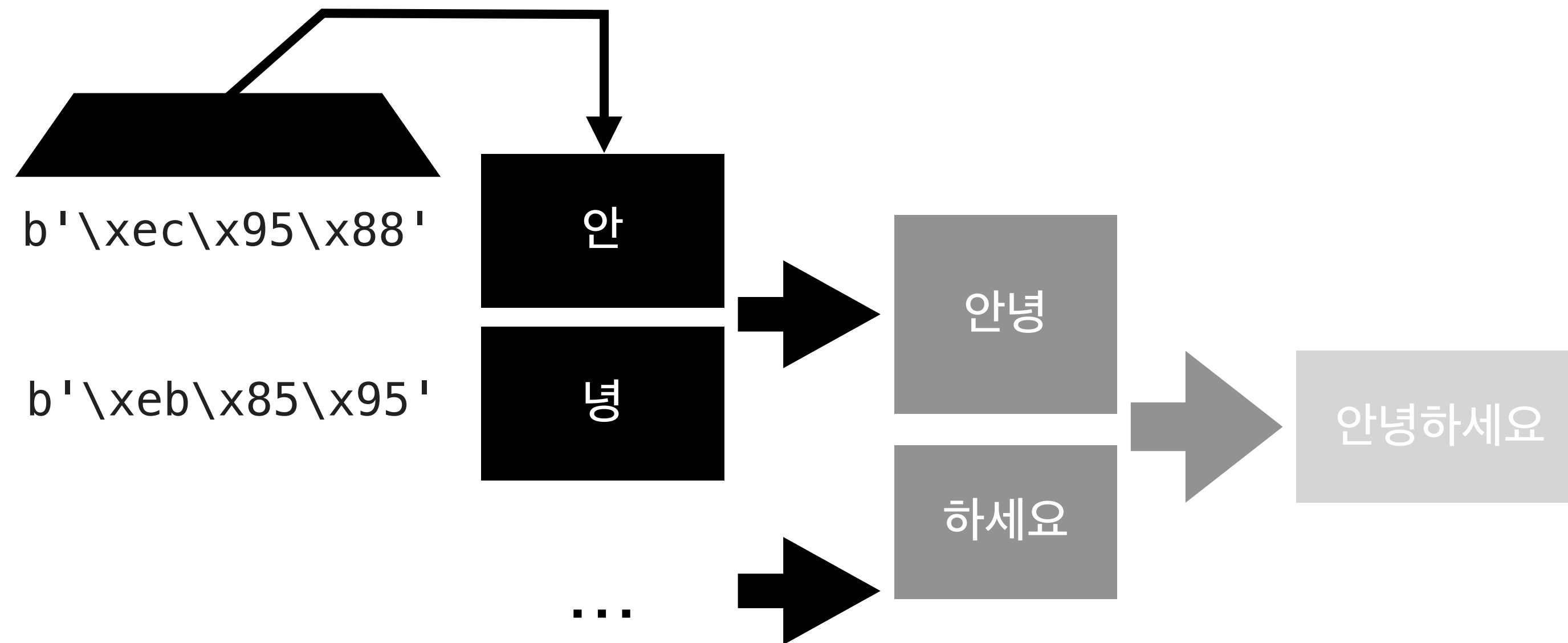
```
[10]: 136
```



```
'0xae0x8b0xf80x000x160x7e0xcc0xe60x360x69'
```

무작위 숫자도 문자?

문자란 무엇일까? for representing as vector

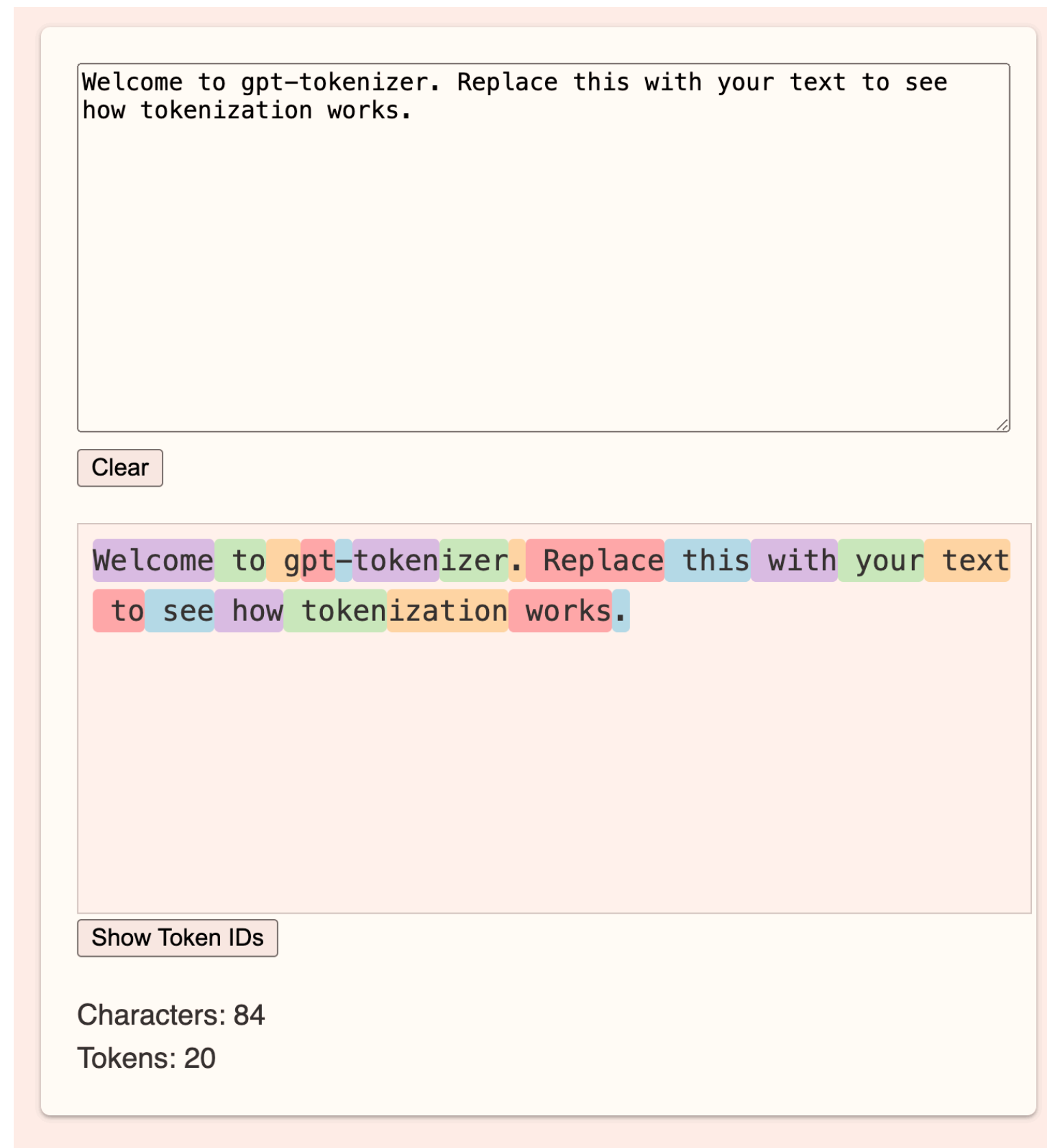


Low-entropy 쌍을 융합

= 응집성 높은 친구들의 합성

= 문자 ~~ 압축

문자란 무엇일까? for representing as vector



Low-entropy 쌍을 융합

= 응집성 높은 친구들의 합성

= 문자 ~~ 압축

Tokenizing... : 문자->숫자

a:1, b:2, c:3, ...

(1, 2)=3

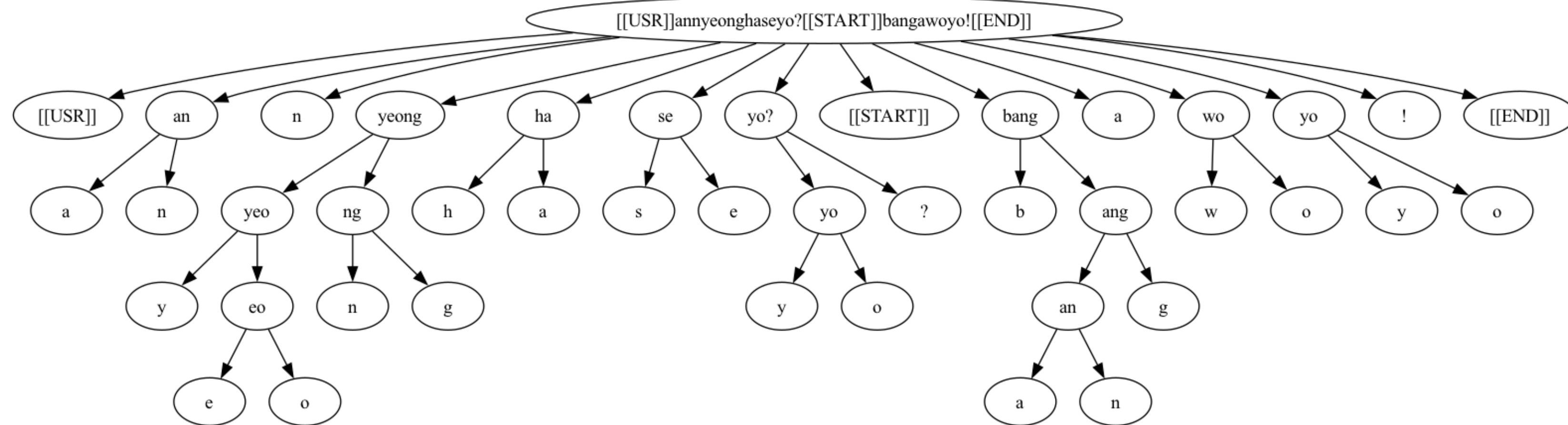
숫자

(3, 1)=4

페어정보

...

BPE, senetencepiece...



Embedding... : 문자->숫자->dense vector

1	<p>	0
2	<unk>	0
3	<s>	0
4	</s>	0
5	_이	-0
6	_있	-1
7	_그	-2
8	으로	-3
9	에서	-4
10	_사	-5
11	_수	-6
12	_아	-7
13	_하	-8
14	_것	-9
15	_대	-10
16	_1	-11
17	_가	-12
18	_한	-13
19	_자	-14
20	_보	-15
21	_시	-16
22	_다	-17
23	_2	-18
24	_기	-19
25	_나	-20
26	_전	-21
27	지	-22

[0, 2, 6, 3, 1, 5, ...]

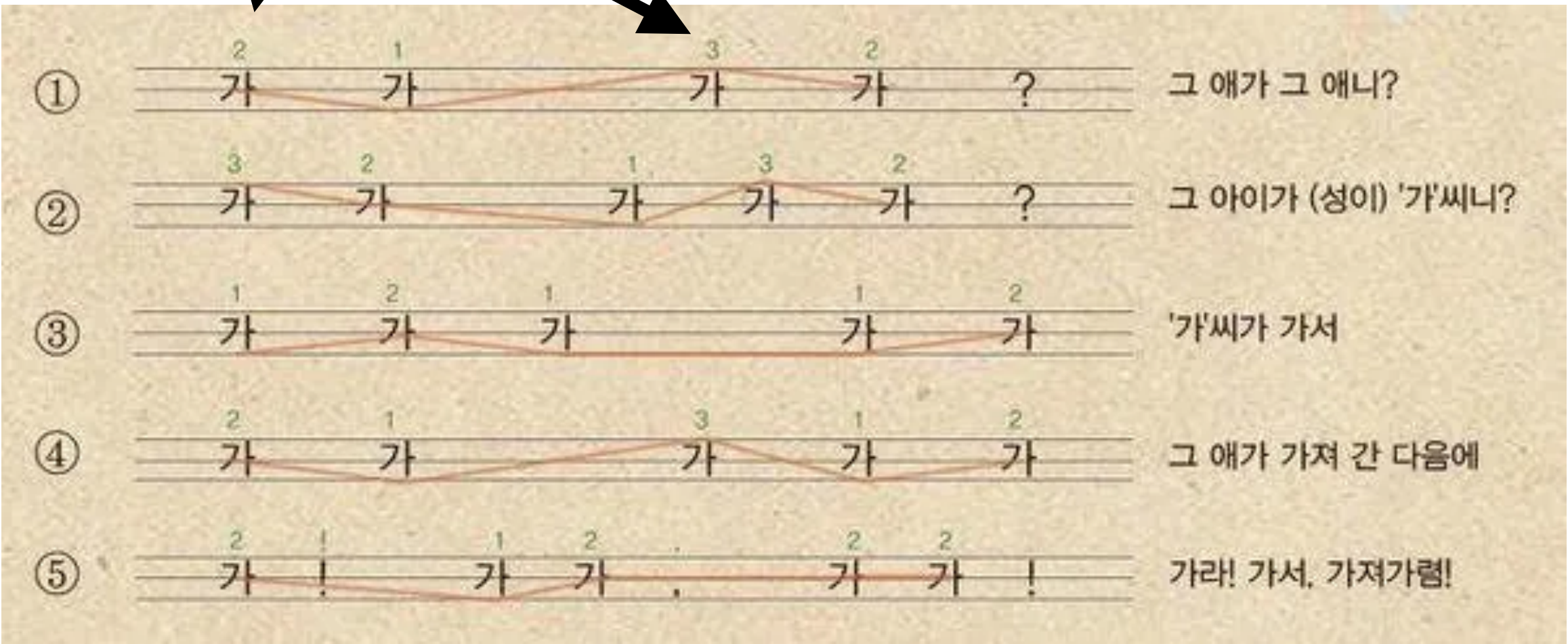
$W \sim R(n_token, d_{emb})$

$emb = W[n, :]$

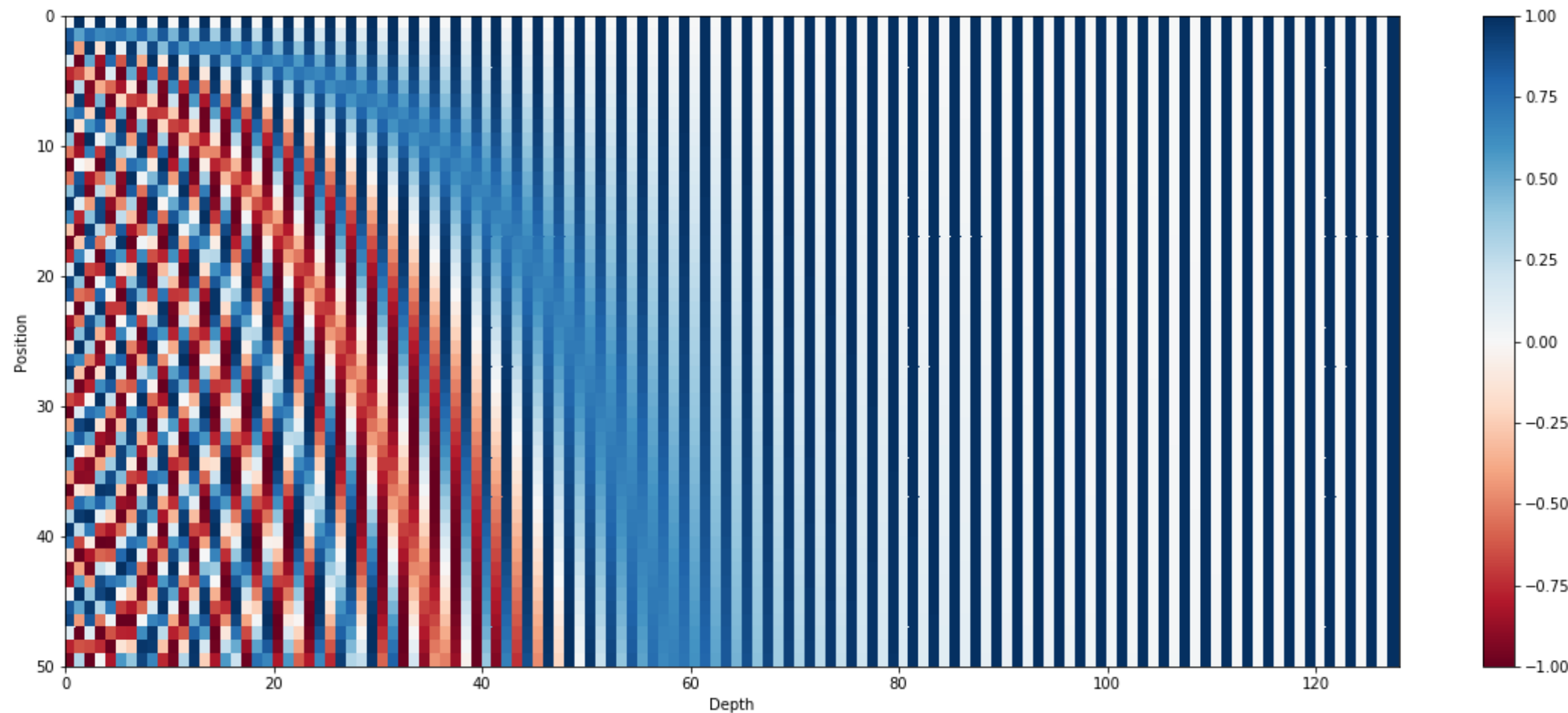
Pos Embedding... : 당신은 어디에 위치합니다.

1	<pad>	0
2	<unk>	0
3	<s>	0
4	</s>	0
5	_이	-0
6	_있	-1
7	_그	-2
8	으로	-3
9	에서	-4
10	_사	-5
11	_수	-6
12	_아	-7
13	_하	-8
14	_것	-9
15	_대	-10
16	_1	-11
17	_가	-12
18	_한	-13
19	_자	-14
20	_보	-15
21	_시	-16
22	_다	-17
23	_2	-18
24	_기	-19
25	_나	-20
26	_전	-21
27	지	-22

[0, 2, 6, 3, 1, 5, ...]



Pos Embedding... : 당신은 어디에 위치합니다.



Explicit Positional Embedding

For a position pos and embedding dimension i :

- $PE_{(pos, 2i)} = \sin \left(\frac{pos}{10000^{\frac{2i}{d_{model}}}} \right)$
- $PE_{(pos, 2i+1)} = \cos \left(\frac{pos}{10000^{\frac{2i}{d_{model}}}} \right)$

Learnable Positional Embedding

아까의 embedding layer에서
token 입력 대신 위치(t)!

Pos Embedding... : 당신은 어디에 위치합니다.

1	< >	0
2	<unk>	0
3	<s>	0
4	</s>	0
5	_이	-0
6	_있	-1
7	_그	-2
8	으로	-3
9	에서	-4
10	_사	-5
11	_수	-6
12	_아	-7
13	_하	-8
14	_것	-9
15	_대	-10
16	_1	-11
17	_가	-12
18	_한	-13
19	_자	-14
20	_보	-15
21	_시	-16
22	_다	-17
23	_2	-18
24	_기	-19
25	_나	-20
26	_전	-21
27	지	-22



[0, 2, 6, 3, 1, 5, ...]

+posW[0]

+posW[1]

+posW[2]

가

가

가

+) 음성에 대한 정보를 줄수도!/\...

Attention is all you need



A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.

뭘 보고 말했구나~

= 너가 왜 작동하구나

= 어떻게 작동하는지 알겠다!

Show, Attend and Tell: Neural Image Caption Generation

Self-Attention(q, k, v)

$$\text{softmax}\left(\frac{\begin{matrix} \text{Q} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{K}^T \\ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \end{matrix}}{\sqrt{d_k}}\right) \begin{matrix} \text{V} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \end{matrix}$$
$$= \begin{matrix} \text{Z} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \end{matrix}$$

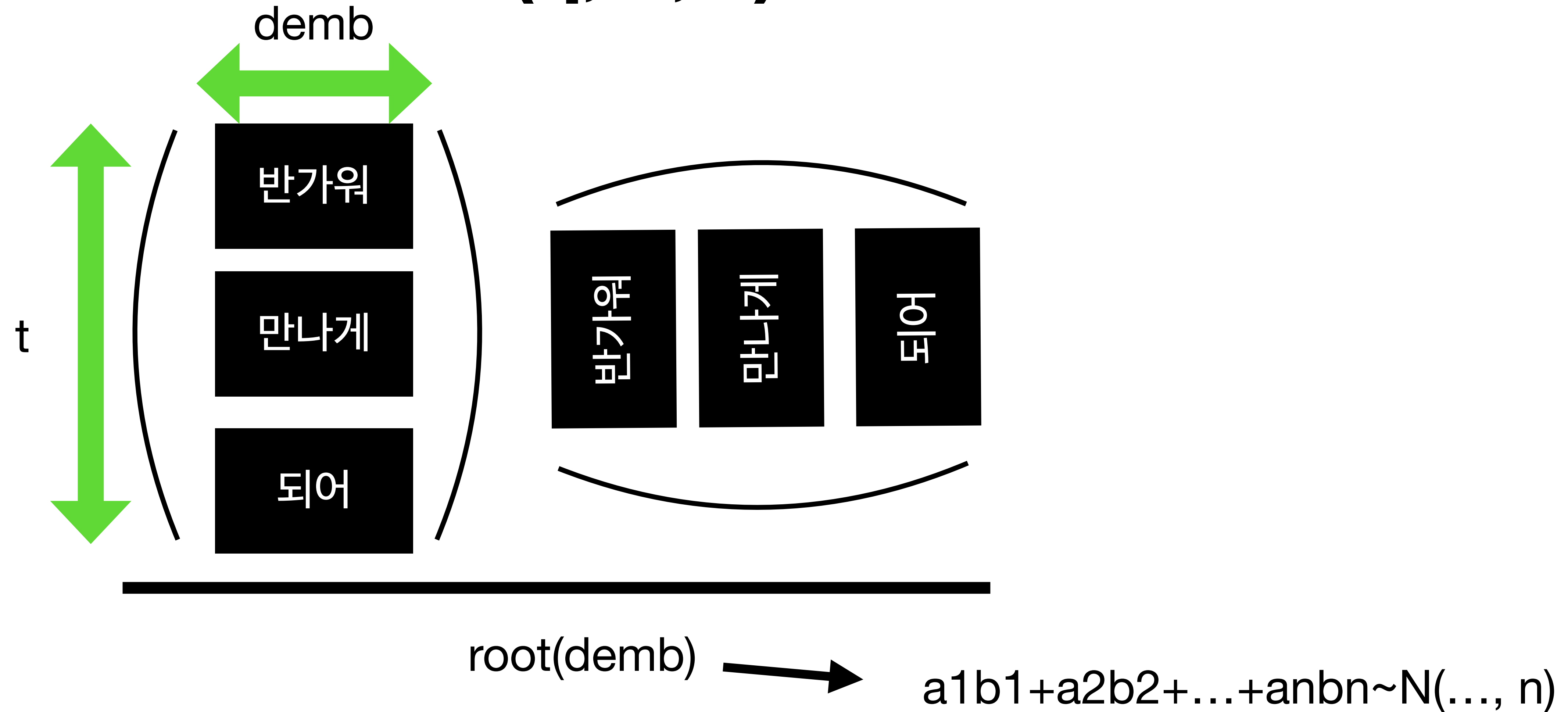
q, k, v from same source

= self-attention

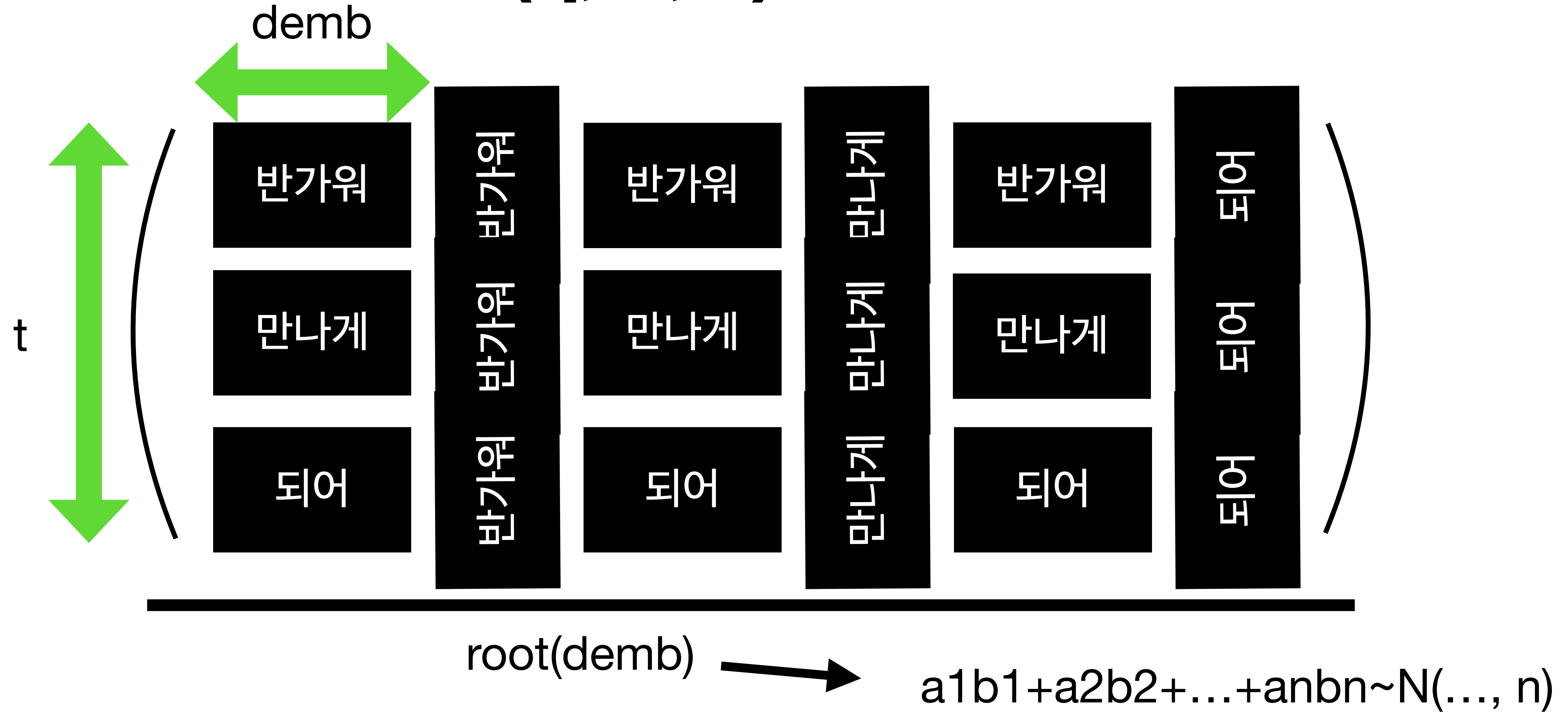
$\sim R(T, \text{demb})$

T: 시퀀스 길이, demb: 임베딩 차원

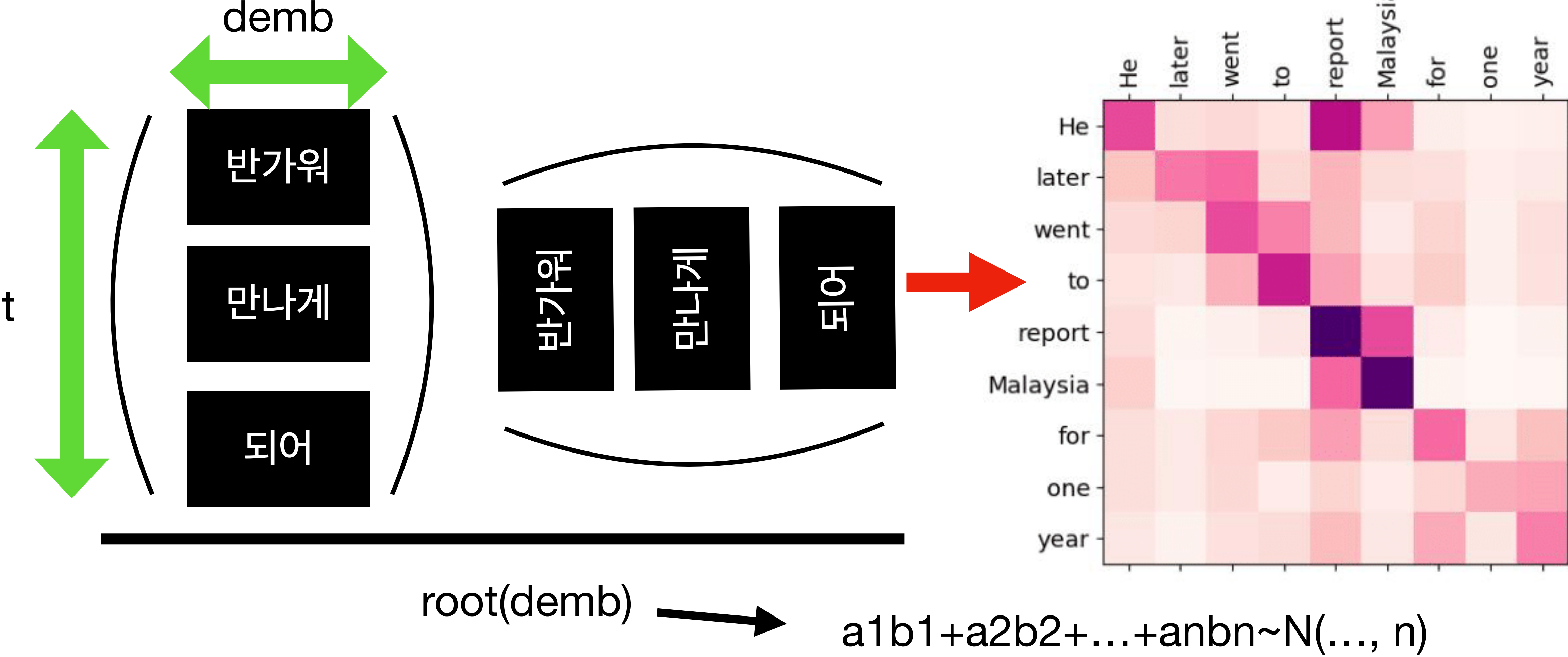
Self-Attention(q, k, v)



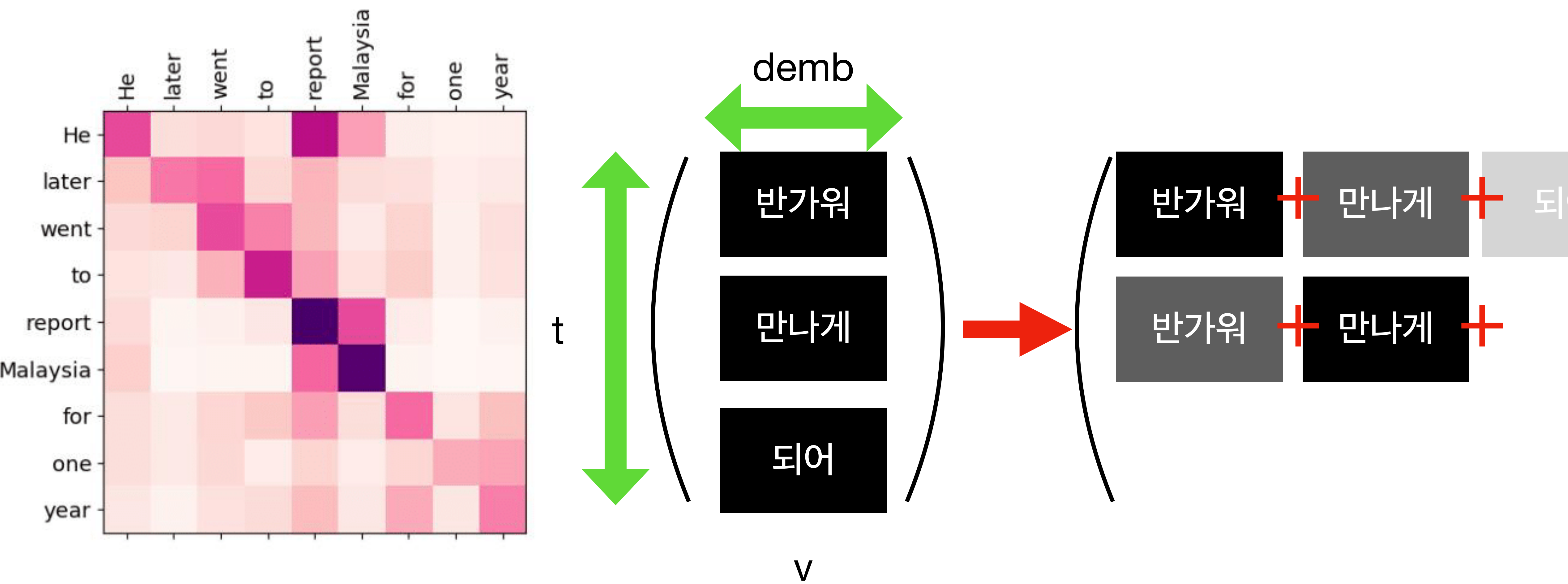
Self-Attention(q, k, v)



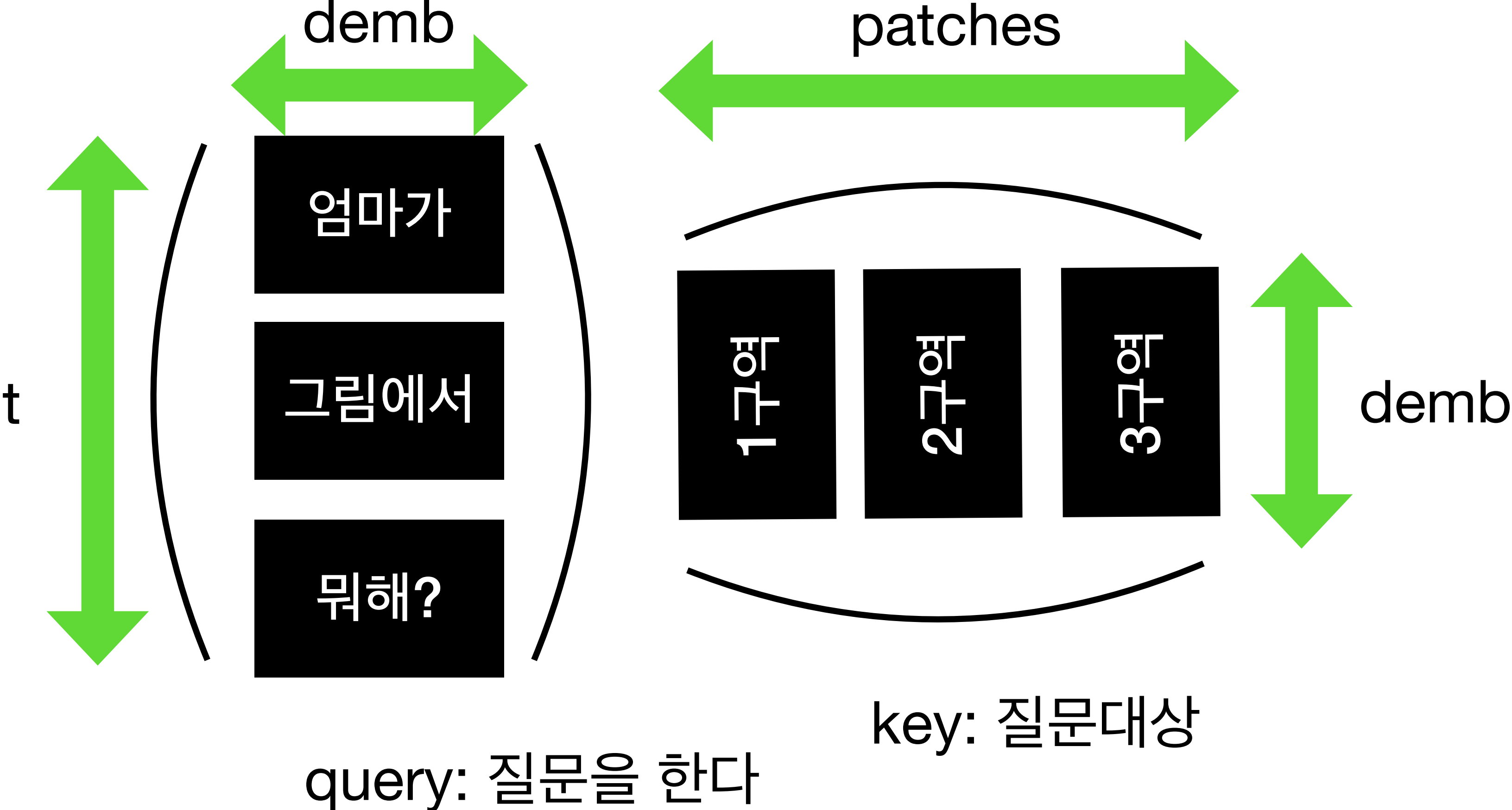
Self-Attention(q, k, v)



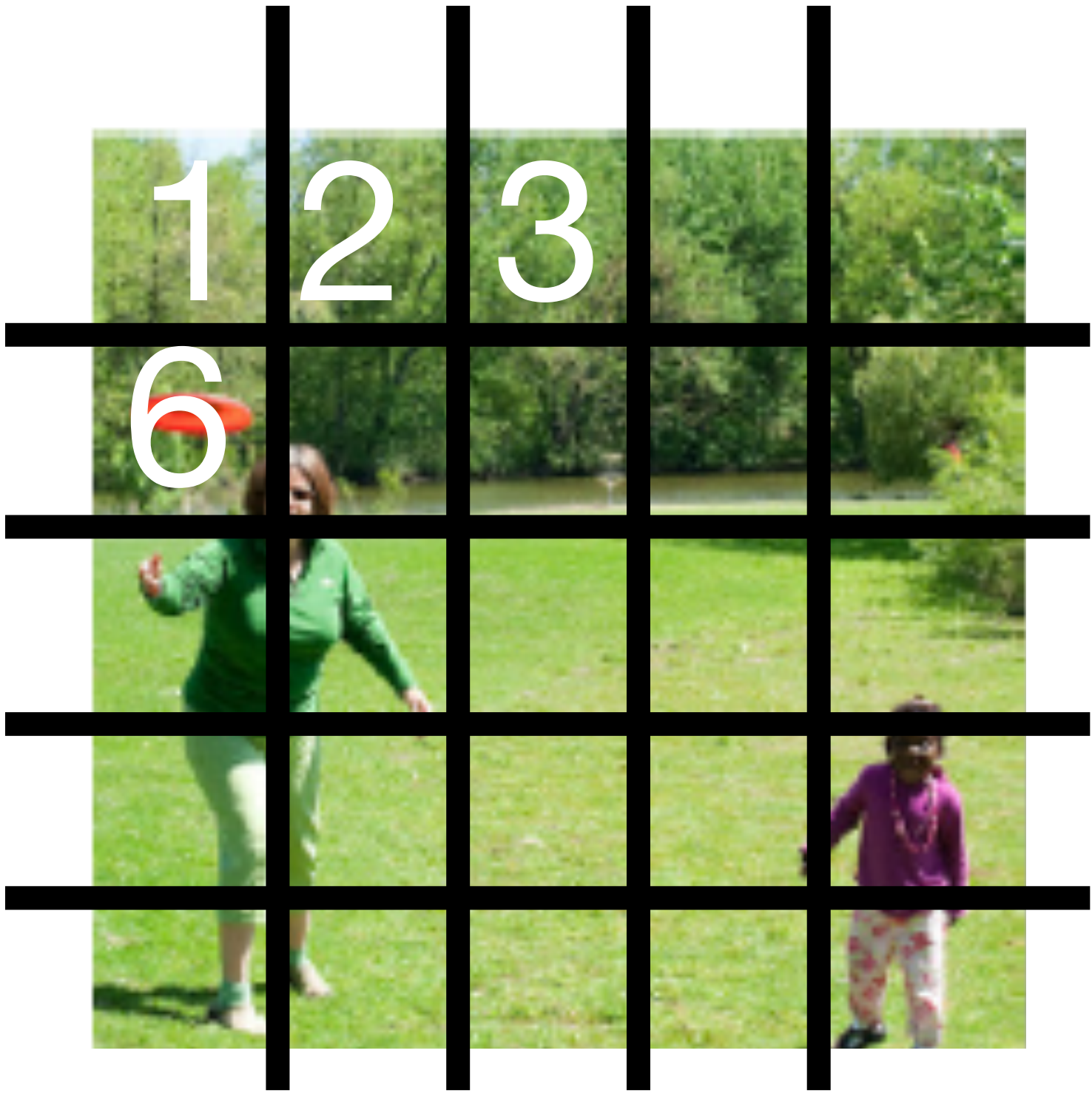
Self-Attention(q, k, v)



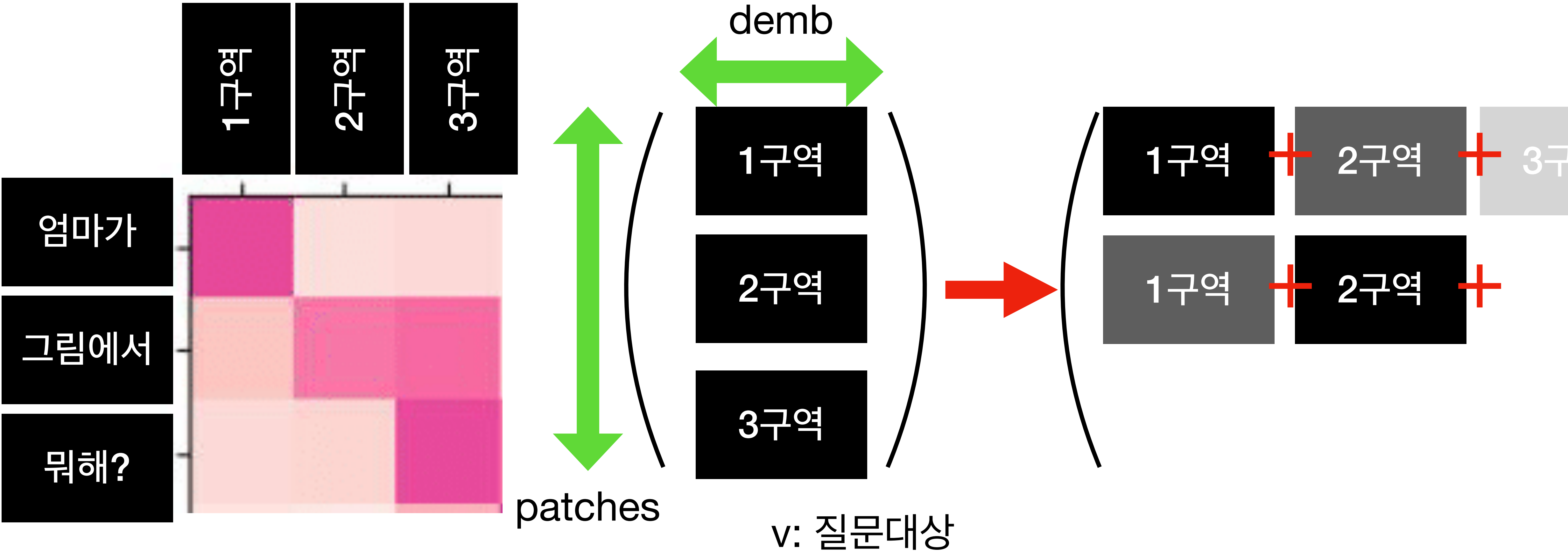
Cross-Attention(q, k, v)



google.com/search?q=질문



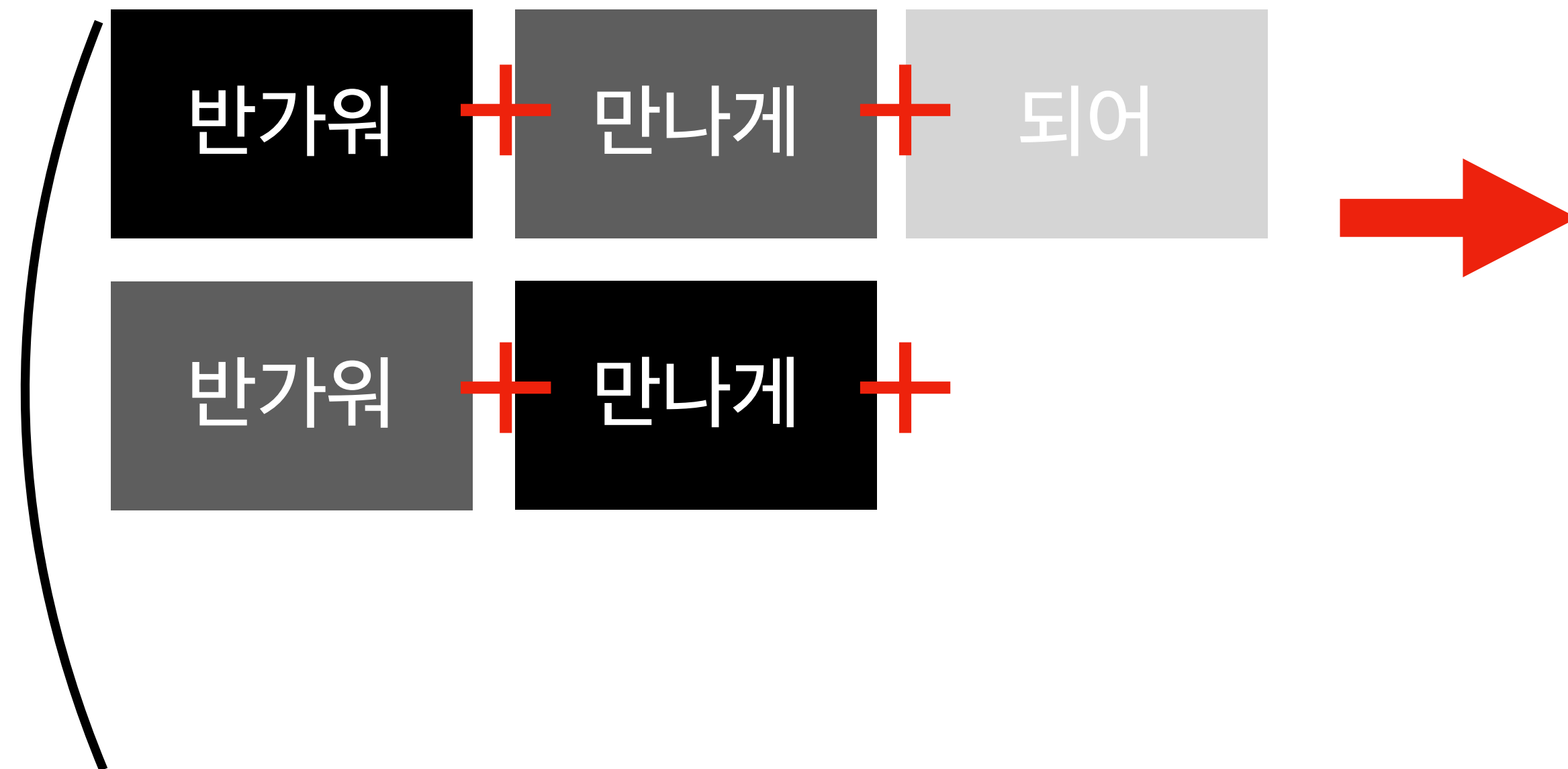
Cross-Attention(q, k, v)



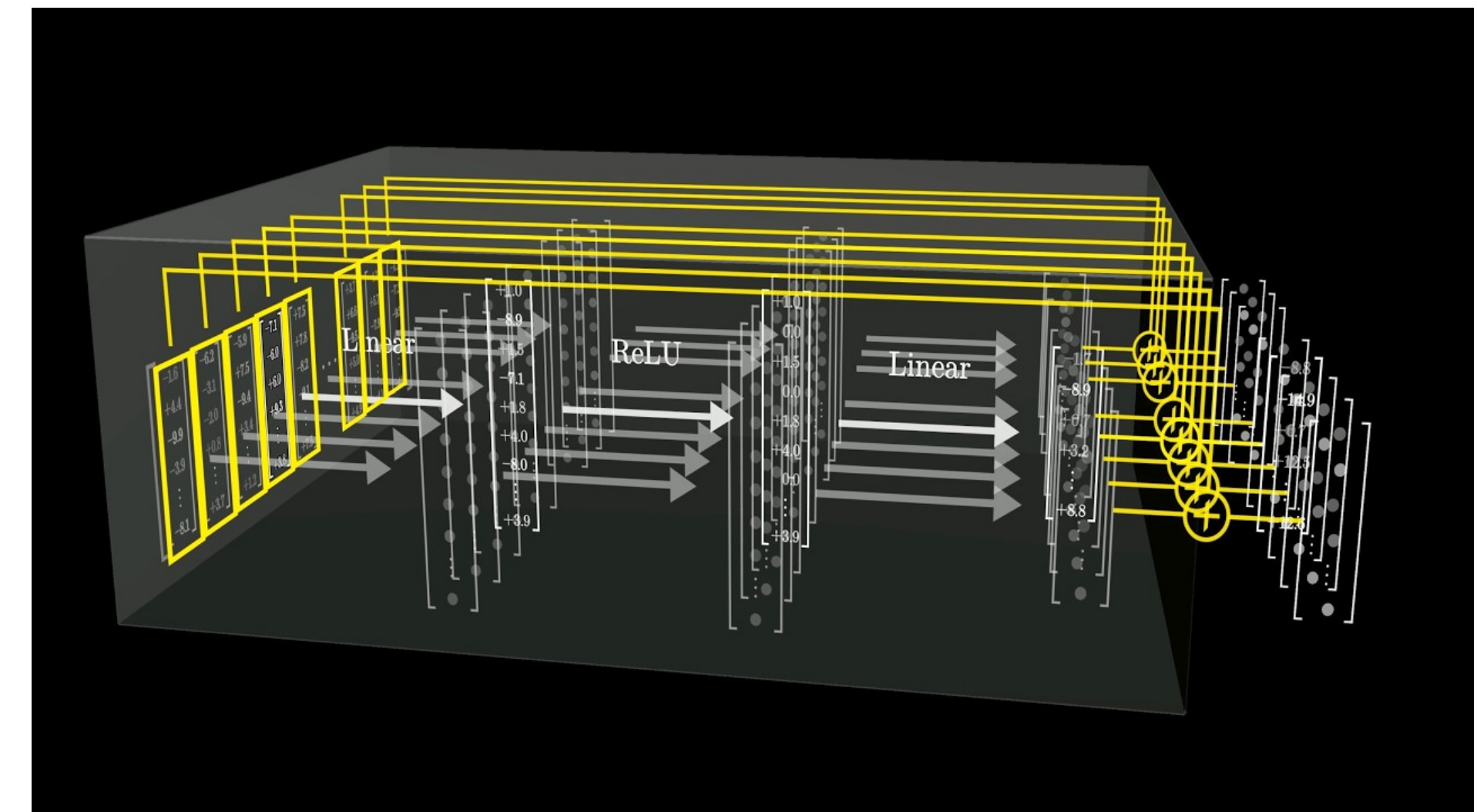
Linear Projection

$W1 \sim R(demb, 3 \times demb)$

$W2 \sim R(3 \times demb, demb)$



How might LLMs store facts | DL7 by #b1B



Attention 연산은 Learnable Parameter의 부족

- 데이터에 내재된 정보의 극대화

Linear Projection

$W1 \sim R(demb, 3 \times demb)$

$W2 \sim R(3 \times demb, demb)$

왜 3일까?

차원 확장 $\rightarrow$ 축소 why??

도대체 왜지 감자?

Linear Projection

$W1 \sim R(demb, 3xdemb)$

$W2 \sim R(3xdemb, demb)$

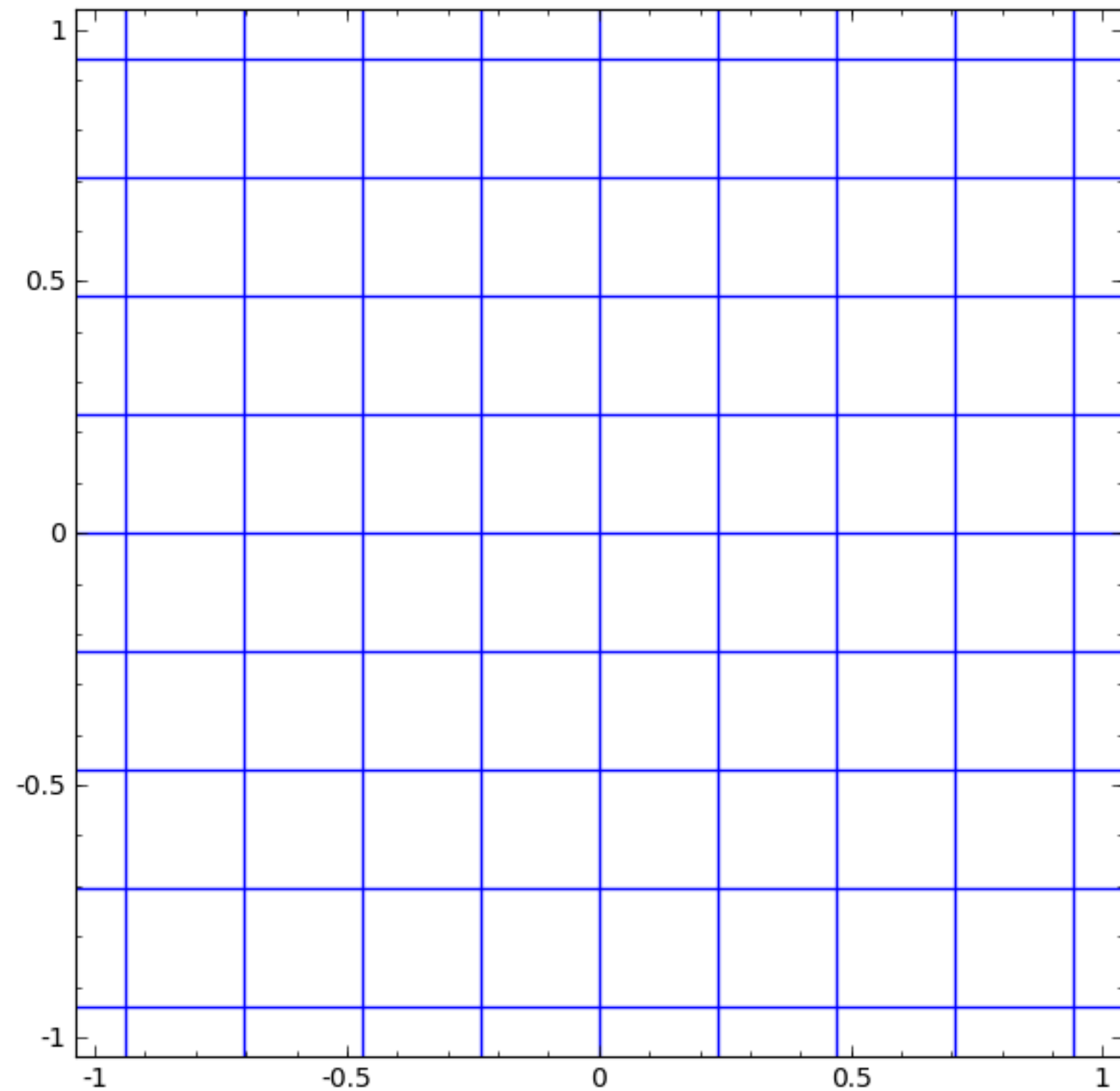
$2n+2 < 3n$???인가???

Links and knots are 1-dimensional manifolds, but we need 4 dimensions to be able to untangle all of them. Similarly, one can need yet higher dimensional space to be able to unknot n -dimensional manifolds. All n -dimensional manifolds can be untangled in $2n + 2$ dimensions.⁶

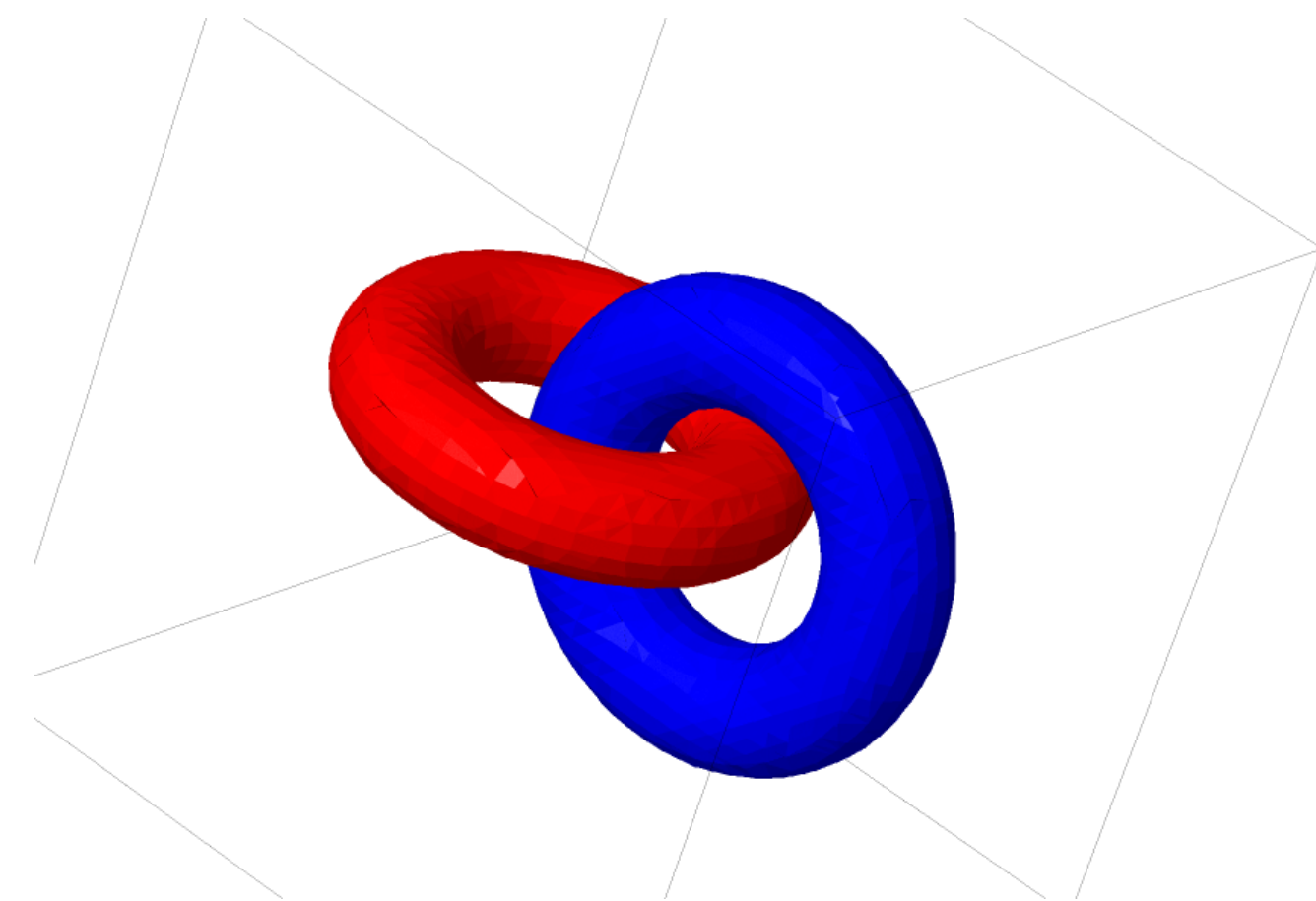
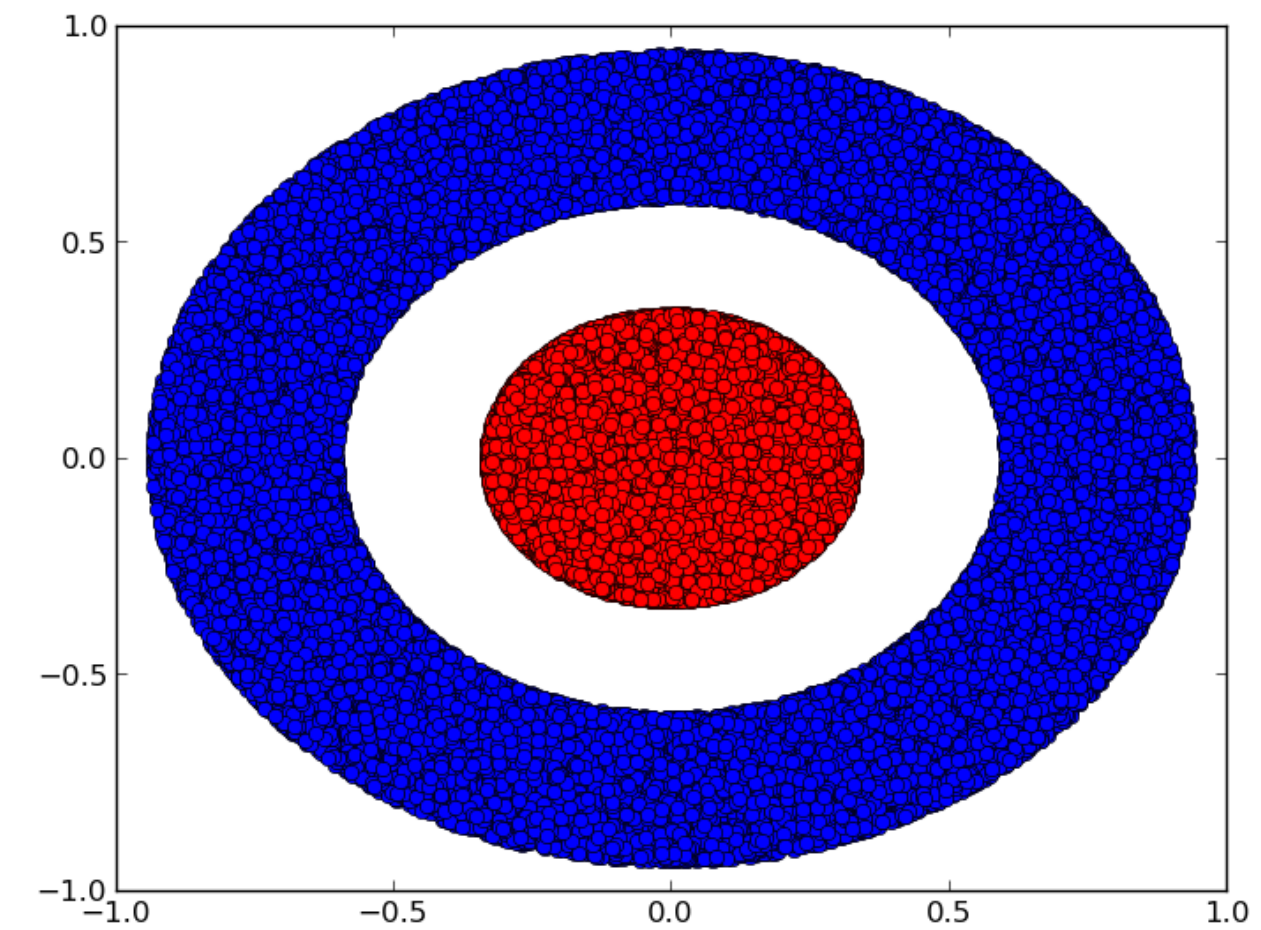
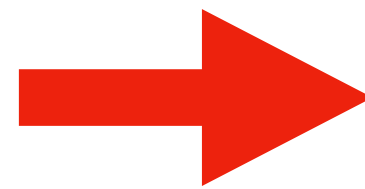
Neural Networks, Manifolds, and Topology:

<https://colah.github.io/posts/2014-03-NN-Manifolds-Topology/>

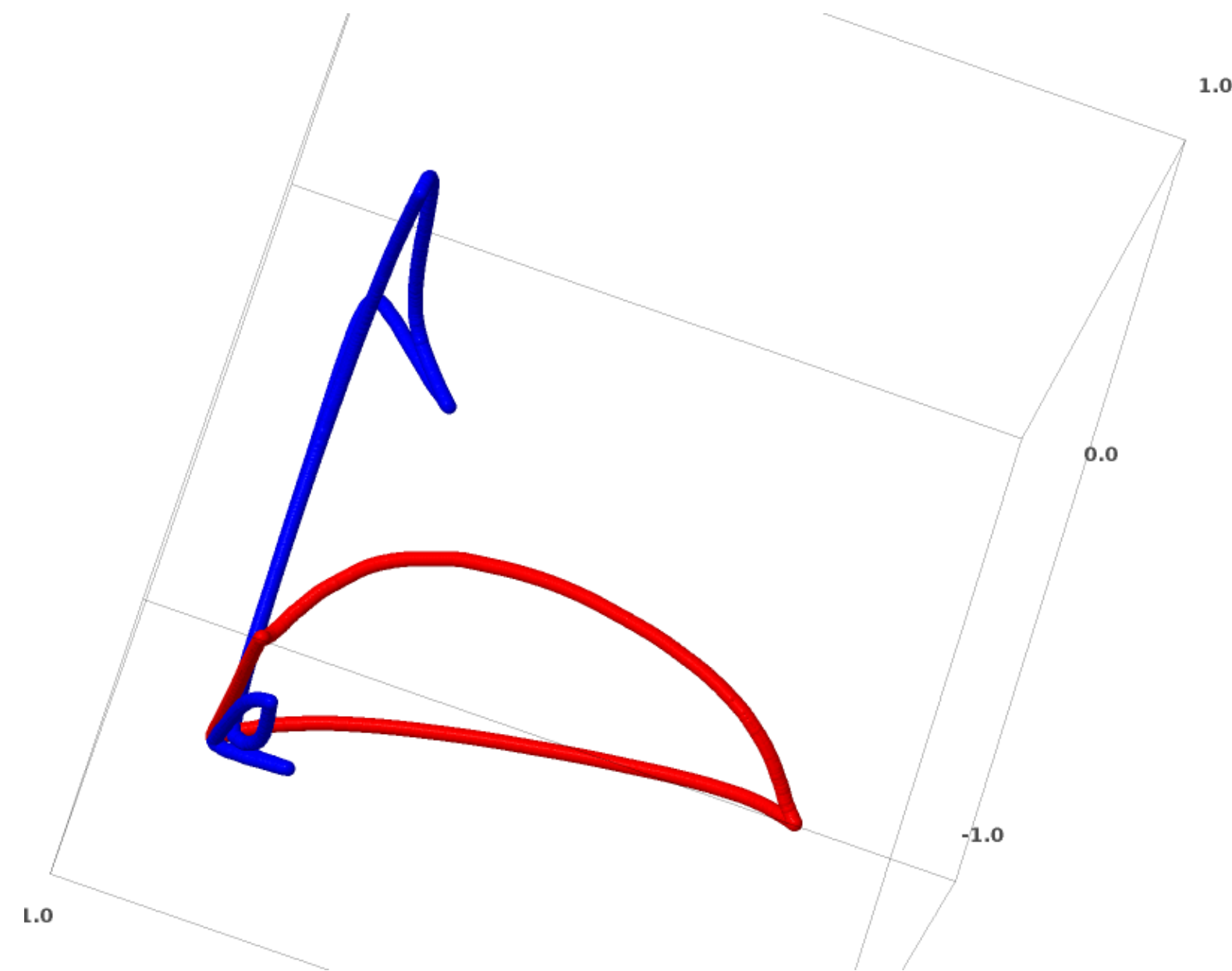
Linear Projection



연속적인 변환을 통해서..



Linear Projection



결국 변환을 거친 representation의
선긋기를 통해서 판단하기 때문에
오차율 달성 0이 knot의 여부를?



Linear Projection

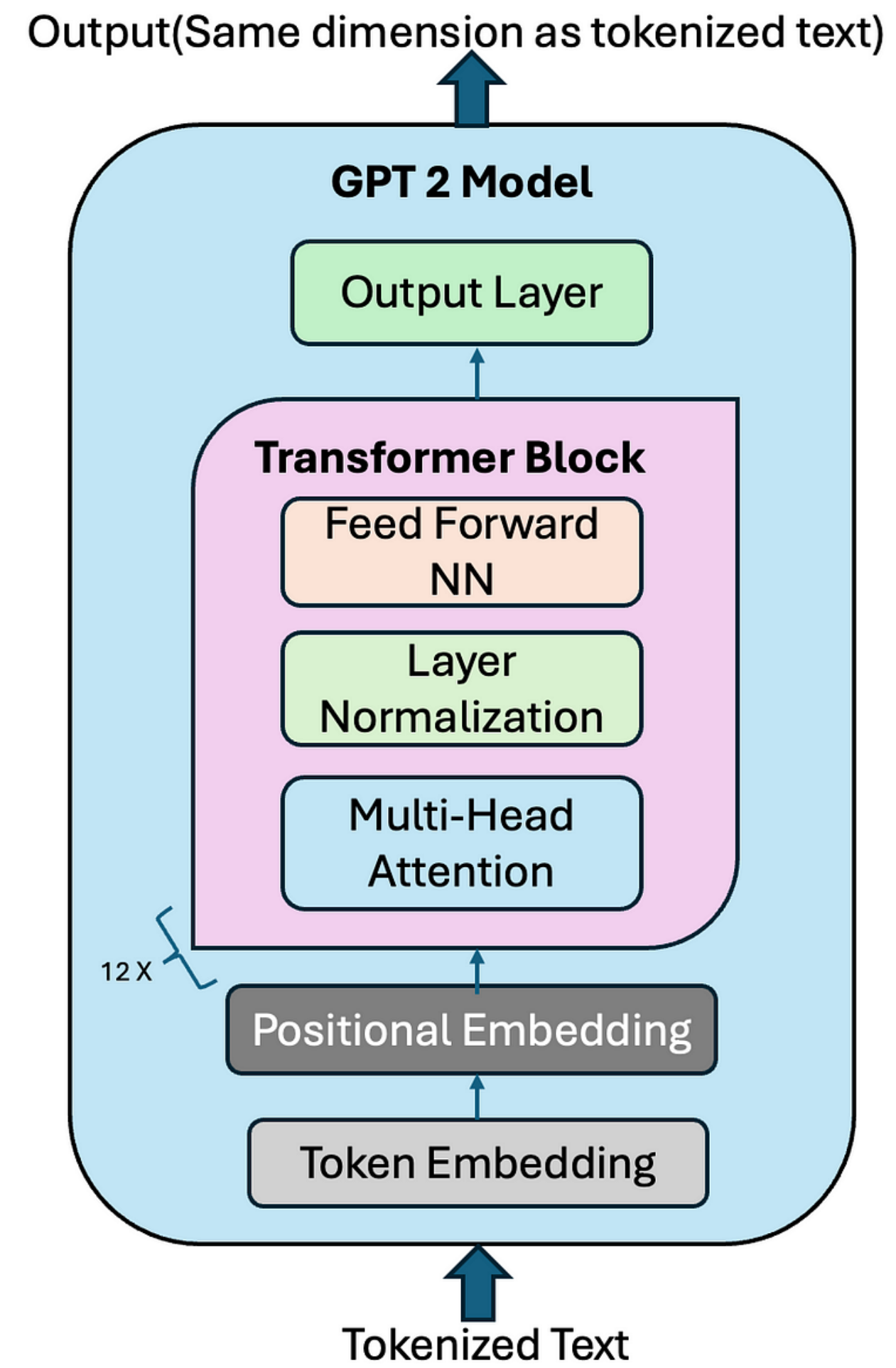
알 수 없는 모양을 가진 아주 복잡한 데이터는 결국 n 차원 다양체

n 차원 다양체를 unfold하기 위해서

$2n+2 < 3n$ (안전용?)

-> whitney embedding theorem?(위상수학 공부핍 구합니다.)

끝!



번외(before Deepseek-알1)

- Rotary Positional Embedding(RoPE): 직접 더할필요?
- Grouped Query Attention(GQA): MHA too costly
- layer normalization
- beyond vanilla attention: n^2 무서워요..

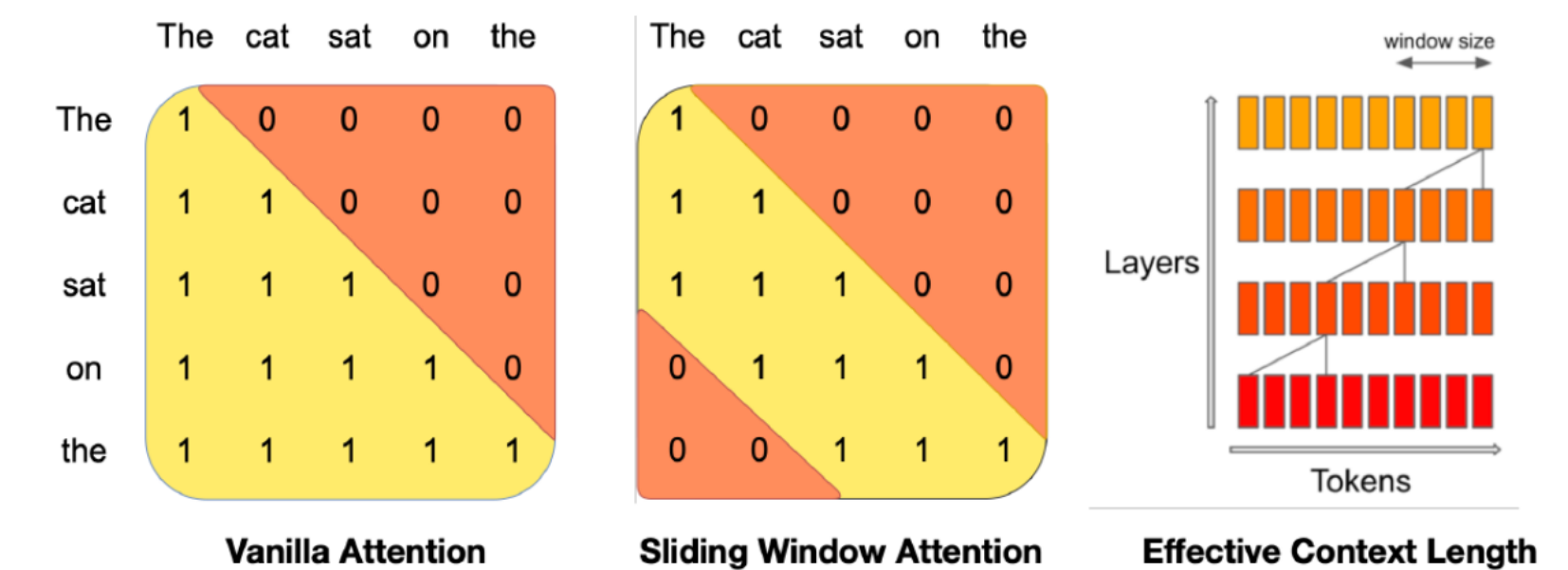


Figure 1: Sliding Window Attention. The number of operations in vanilla attention is quadratic in the sequence length, and the memory increases linearly with the number of tokens. At inference time, this incurs higher latency and smaller throughput due to reduced cache availability. To alleviate this issue, we use sliding window attention: each token can attend to at most W tokens from the previous layer (here, $W = 3$). Note that tokens outside the sliding window still influence next word prediction. At each attention layer, information can move forward by W tokens. Hence, after k attention layers, information can move forward by up to $k \times W$ tokens.

과제

과제 making in progress..

다음주 월요일(4/7)에 예시 모델과 벤치마크 사용법 올리겠습니다.

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시험기간동안(~4/28)은 과제를 해결하는 것으로 부담을 경감(?)하려고 합니다.

각자 열심히 공부, 공부하고 만납시다~!

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이후에는 조별로 딥씩 이후 + 강화학습 + ICL에 대해 스터디 예정입니다